BCG FED

Centers for Medicare & Medicaid Services (CMS) Request for Information (RFI) | Health Technology Ecosystem

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Introduction

BCG applauds CMS and HHS for seeking public input on this important initiative to improve health technology and empower Medicare beneficiaries and caregivers. We welcome the agency's commitment to engaging a broad range of stakeholders and are pleased to contribute perspectives on three of the key questions outlined in the Request for Information (RFI). We hope our input supports CMS in designing forward-looking strategies that enhance digital access and data interoperability, strengthen value-based care, foster innovation, and support cost compression across the Medicare and broader healthcare ecosystem.

BCG response to select RFI questions

Patients & Caregivers # PC-12: What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

- ***a. Examples may include the following:***
  - ***1) Binding cost estimates for pre-defined periods.***
  - ***2) Viewing provider schedule availability.***
  - ***3) Using third-party apps for appointment management.***
  - ***4) Accessing patient-facing quality metrics.***
  - ***5) Finding the right provider for specific healthcare needs.***
- ***b. What use cases are possible today?***
- ***c. What should be possible in the near future?***
- ***d. What would be very valuable but may be very hard to achieve?***

Generative AI and digital health innovations are actively addressing a range of key patient and caregiver use cases, such as care navigation and access to provider services. From BCG's experience serving leaders and innovators in the healthcare industry, we know AI chatbots and virtual assistants are already used to triage patient needs and deliver health guidance, with many early adopters taking "stepping stone" approaches to test new capabilities – like AI co-pilots for "nurse line" staff – that support greater patient participation in their care journey while maintaining human oversight and quality assurance. Additionally, at-home diagnostic solutions are streamlining appointment logistics by enabling remote diagnostics and asynchronous workflows, benefitting harder to reach patient populations, such as those in rural settings. Smart implants and continuous monitoring devices are also advancing chronic disease management, giving patients more control while feeding critical data to caregivers for proactive interventions.

Looking ahead, more sophisticated digital ecosystems are expected to automate entire patient episodes—from intake through treatment—enhancing both care coordination and operational efficiency. Interoperable systems and secure data-sharing frameworks will continue to be critical to allow for seamless data exchange across providers, improving transparency and enabling the delivery of real-time, personalized insights to patients and caregivers alike. Meanwhile, future innovations like vocal biomarkers and AI-driven diagnostics offer promise for early detection and tailored care. To drive these advancements, it will be critical that the healthcare industry works together with CMS to effectively mitigate potential risks – biased outputs, false and opaque results, privacy concerns, and overreliance – and create an environment of regulatory clarity that encourages both industry innovation and patient trust. By harnessing these emerging

capabilities, we can shift the health care ecosystem toward proactive, data-driven health systems designed around the needs of both patients and caregivers.

BCG has experience partnering with industry and digital health leaders across the ecosystem to observe and shape how these use cases play out in real time. Please see several perspectives on this topic in Appendix A (*How Digital and AI Will Reshape Health Care in 2025*) and Appendix B (*Generative AI Will Transform Health Care Sooner Than You Think*).

Technology Vendors, Data Providers, and Networks #TD-1: What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

While there are important policy and regulatory levers CMS can apply to shape the broader digital health ecosystem, a complementary – and similarly critical – set of opportunities lies in how CMS manages its own role as a buyer and partner in developing digital health products for Medicare beneficiaries and caregivers. CMS can further modernize its internal approach to digital product development by adopting industry vendor management best practices that emphasize engineering efficiency, agility, transparency, and accountability. One of the most powerful tools in this regard is the establishment of a dedicated, independent product control tower-like or "navigator" function – an objective third party composed of capable product managers, IT architects, program-specific subject matter experts, and user-experience designers. This capability enables an agency such as CMS to maximize its return on investment (ROI) and drive operational cost compression by ensuring that product requirements align with the agency's strategy, are coordinated across components, and mitigate waste from cost overruns due to misalignment. A skilled partner can serve as a trusted advisor to CMS and as a liaison with vendors, bringing the right stakeholders to the right conversations at the right time to accelerate better outcomes from the start of a transformation, while in parallel helping to build capabilities from within for the agency to effectively manage ongoing delivery. Using this flexible approach, the navigator function can serve as a cost-effective bridge between CMS and its vendors, helping to define business cases, shape the roadmap, and support vendor negotiations and management for digital health products (and even more broadly).

In our digital health transformation experience across public and private sectors, navigator-led transformations result in significant improvements in project ROI, benefitting the taxpayer through efficiency of federal dollars, timeliness, reduced waste, and improved government services. Benefits include:

- 40% faster path to value from new capabilities
- 35% reduction of project scope
- 70% increased degree of standardization
- 35% higher transformation success rate
- 30- to 60-fold return from investing 5% of program budget in de-risking

This approach aligns with the evolution toward platform-based ecosystems, where value is created through network effects and the exchange of products and services for mutual benefit. By adopting the role of orchestrator, CMS can coordinate a broad set of participants—vendors, caregivers, developers, and beneficiaries—into a cohesive and collaborative whole. This enables

CMS to combine aggregated supply and demand with high service levels and reduced friction, creating the conditions for new multiparty solutions to emerge. Acting as a central node, CMS can stimulate network effects by increasing the diversity and relevance of offerings while maintaining best-in-class functionality, tailored user experiences, and transparent interfaces that reduce barriers to participation. As with successful orchestrators in the private sector, CMS has the opportunity to expand the impact of its digital efforts beyond individual transactions – supporting a dynamic ecosystem that scales innovation, enhances partner contributions, and delivers greater value to beneficiaries over time.

More on the strategic importance of the product control tower / navigator concept and other key steps to deliver effective and efficient digital transformations for government agencies like CMS can be found in a recent BCG publication, please see Appendix C (*How Governments Can Keep Digital Transformations on Time and on Budget*). For additional perspectives on private sector exemplars for platform-based ecosystem innovations, please see Appendix D (*The Rise Of AI, Platforms, And Ecosystems*).

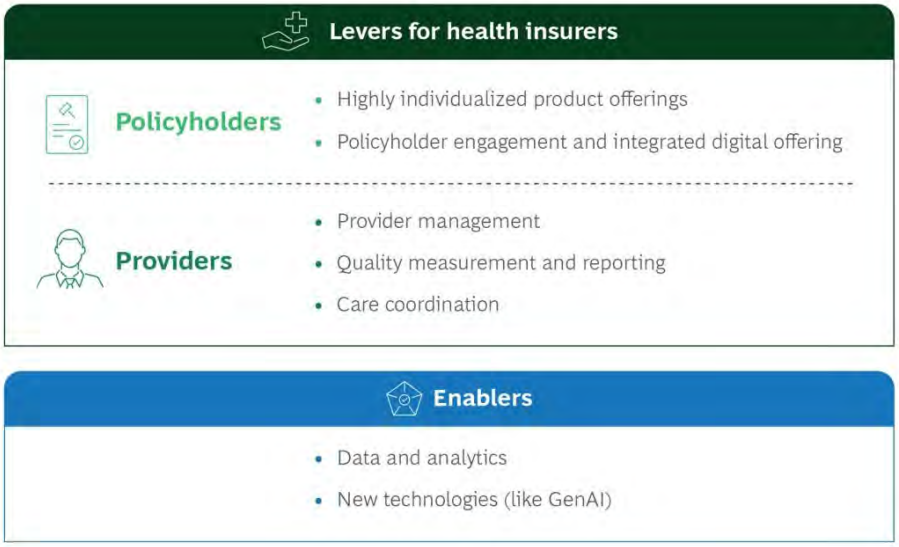
Value Based Care Organizations # VB-3: What are essential health IT capabilities for value-based care arrangements?

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

We believe that IT capabilities – particularly data & analytics and new technologies such as GenAI – are critical enablers for health payers and providers developing value-based care arrangements (see Figure 1 below).

Figure 1. A Value-Based Care Framework for Health Insurers



Source: BCG analysis.

Health care payers and providers must strengthen their technology foundations to successfully advance value-based health care arrangements. Building a modern platform that enables real-time, cross-functional insight is no longer optional. Healthcare organizations can develop an augmented set of dashboards that use AI to generate actionable, cross-cutting insights across programs and populations. These tools are essential for moving away from siloed, reactive operations and toward more proactive, data-driven strategies that enable faster decision-making and better outcomes. They empower organizations to gain greater control over cost and outcome variability, help them to identify inefficiencies, and translate insights into actionable measures that guide the design and execution of transformational value-based strategies.

Investment in emerging technologies can also directly enhance the value that patients experience through the health system. New technologies like GenAI now make it feasible for payers and providers to personalize communication and engagement with patients at scale in a way that can help empower people to achieve their health goals and align financial incentives with health – something previously prohibitively expensive. By prioritizing digital channels that address patient pain points and provide personalized experiences, organizations have the opportunity to improve access, engagement, and satisfaction. Additionally, by applying advanced analytics to support robust provider management, insurers can negotiate rates more efficiently, design treatment bundles, and establish clear governance to deliver the highest quality clinical care at an acceptable cost. User-friendly digital platforms can also streamline case management and improve outcomes by connecting patients and providers, enhancing the understanding of patient needs and offering real-time support throughout the care journey.

For more information, please see Appendix E (*Redefining Global Health Insurance Through Value-Based Care*).

Appendices: Referenced BCG articles

Appendix A: How Digital and AI Will Reshape Health Care in 2025

Appendix B: Generative AI Will Transform Health Care Sooner Than You Think

Appendix C: How Governments Can Keep Digital Transformations on Time and on Budget

Appendix D: The Rise Of AI, Platforms, And Ecosystems

Appendix E: Redefining Global Health Insurance Through Value-Based Care

How Digital and AI Will Reshape Health Care in 2025

The definition of digital health is evolving. The era spurred on by the Covid-19 pandemic—think telemedicine and digital therapeutics, which have struggled to scale—is giving way to one defined by artificial intelligence (AI) and solutions that strengthen the bond between health care professionals and patients in an integrated manner, with appropriate economics to support them.

We see this shift reflected in trends that experts across BCG and BCG X anticipate will shape digital health in 2025. As AI matures, it is rapidly expanding possibilities for patients, providers, and health care organizations alike. New digital solutions are being leveraged to address gaps in care for chronic conditions such as heart failure, diabetes, and mental health. And the growing influence of generative AI (GenAI) on every aspect of health care—from personalized care to automated workflows—is a key theme for the upcoming year, as it was in 2024.

Let's dive deeper into how we expect digital and AI solutions to reshape health care in 2025.

Patient Support

This year, digital health tools will continue to transform patient care, improving their support and access. Smart implants and wearable devices that allow providers to monitor patients' cardiac activity, blood sugar levels, and other biological functions in real time from remote locations will enable better chronic disease management and improve patients' quality of life. As sleep continues to gain attention as a crucial biomarker for overall well-being, health tech companies are creating more advanced, accurate sleep-tracking tools.

A growing number of individually tailored apps and digital platforms will give patients more control over their medical conditions, predict flare-ups, and suggest real-time interventions. We expect consumers to increasingly rely on AI chatbots and virtual assistants for answers to health questions.

Digital health will continue to offer solutions to address gaps in women's health care, including femtech innovations to redesign traditional "hardware" used for women's health (such as the speculum), with the female experience at the center. It's a needed shift: A recent BCG X survey found that fewer than half of women respondents across the globe (41%) agreed that there are sufficient services to address their specific health concerns.

We are also beginning to see a maturing of partnerships between femtech health and wellness brands that can lead to interoperable ecosystems that pool women's health data and ultimately drive improved health outcomes.

Provider Empowerment

Providers will be empowered and enabled by digital technology as well. AI can provide the analytical muscle to process vast quantities of personal patient data, powering highly personalized medical treatment tailored to individuals based on their unique health data from continuous monitoring devices, lifestyle inputs, and individual genetics. This enables providers to adjust treatment dynamically based on feedback in real time.

Artificial intelligence decision-making tools will become mainstream in 2025, giving doctors immediate access to evidence-based research and treatment guidelines. GenAI applications will accelerate diagnoses and minimize diagnostic errors, while speeding the delivery of patient care and more accurately predicting patient outcomes.

Emergence of Ecosystems

At the organizational level, our experts anticipate that the coming year will see an expansion of the use of AI to organize and automate entire workflows instead of just specific tasks. For example, rather than an AI tool that facilitates physician note-taking or scheduling, intelligent agents will automate an entire patient episode of care, from intake through treatment plan. Working across departments, AI programs will learn as they go, improving efficiency and outcomes at both the patient and system level. Health systems will benefit, but so will other types of health care organizations such as pharmaceutical companies, where GenAI can transform key activities such as clinical trials and regulatory submissions.

AI-driven data processing will also allow access to data that has until now been considered too disorganized to be useful, such as medical records, clinical notes, and physician/patient interaction information. Clinicians, payers, and drug companies alike will be able to draw out actionable insights from these data sets to improve patient care and outcomes. At the same time, expanded access will enhance different systems' ability to interact with one another, facilitating more seamless collaboration.

While GenAI continues to generate tremendous excitement in the digital health care space, it's not a panacea. Our experts recognize that some of these programs won't deliver anticipated results in 2025. When that happens, we emphasize the importance of going back to the basics: focusing on business outcomes and tracking key performance indicators. In this way, AI failures can drive more focused, sustainable transformation in the long term.

Clearly, 2025 promises to be a transformative year. We're excited to see how AI and more digital solutions reshape health care.

“With AI-driven solutions, wearable devices, and digital triaging tools, patients are no longer passive participants in their care but active managers of their health journey.”

Ashkan Afkhami

Managing Director and Senior Partner

Every patient will soon have the tools to find the right care, support, and treatment tailored to their unique needs. We’re at a tipping point in patient care. With AI-driven platforms, wearable devices, and digital triaging tools, patients are no longer passive participants in their care but active managers of their health journey. Technology is closing critical gaps in care, particularly in underserved communities, by enabling timely guidance, remote consultations, and personalized care plans. These tools empower patients to take charge of their health and promote a seamless, personalized experience that meets them wherever they are—whether at home, in the clinic, or on the go.

Digital health solutions that simplify workflows, optimize resources, and improve patient monitoring will enable clinicians to deliver continuous, high-quality care. AI-assisted technologies are helping to address capacity challenges, reduce diagnostic turnaround times, and improve treatment accuracy. Similarly, decision-support tools and real-time analytics are enabling smarter, safer care delivery. By integrating remote patient monitoring, automation, and predictive analytics, health care professionals can focus on what matters most: treating and supporting patients beyond the clinic walls. This shift will improve efficiency and empower clinicians to deliver proactive, patient-centered care.

By capturing, analyzing, and applying data to drive better treatments, therapies, and operational efficiency, health care organizations will realize their true potential. By leveraging real-time insights and advanced analytics, organizations can detect diseases earlier, streamline care pathways, and optimize operations. Interoperable systems and secure data-sharing frameworks are critical for achieving these outcomes, ensuring data can flow seamlessly across stakeholders. As we move forward, ethical AI frameworks and integrated data strategies will be the catalysts for change, transforming health care into a precision-driven, efficient, and impactful ecosystem.

“Digital tools can help bridge R&D and access gaps, driving equity in health care for women.”

Johanna Benesty

Managing Director and
Senior Partner

Women continue to face barriers to accessing health care. Many factors contribute to lack of access, even in high-income countries—including economic disparities, limited R&D on women-specific health, and systemic biases. For instance, 26% of US women delay care due to cost. For many low-income women, this limits access to essential reproductive health services. Additionally, lack of R&D on women’s conditions like endometriosis or menopause leads to delayed diagnoses and inadequate treatment. Digital health offers solutions to address these gaps. Telehealth has expanded access to underserved areas, and virtual consultations are proven to support women in rural areas who need mental health services. Health apps such as Eve and Flo collect critical health data about women’s menstrual cycles and ovulation, advancing research on women’s health and creating awareness. And AI can help reduce the cost of R&D on target groups. These digital tools can help bridge R&D and access gaps, driving equity in health care for women.

To fully harness digital health’s potential in low- and middle-income countries (LMICs), the health care ecosystem must overcome several challenges. Scaling digital health initiatives effectively remains a primary hurdle because many digital solutions that show promise in pilot stages struggle with long-term sustainability across diverse regions. Investment in digital infrastructure, talent development, and skills training is essential, as health care workers in LMICs often lack the technical training needed to operate and support digital health tools. Moreover, ethical considerations, especially regarding AI use, are paramount. Without clear guidelines and regulatory frameworks, AI risks exacerbating health inequities rather than reducing them. Cultural adaptation and community trust in these technologies are also critical, requiring a user-centered approach that aligns digital health solutions with local values and health care practices. Addressing these challenges will be key to advancing equitable access to health care in LMICs through digital innovation.

“As smart implants become more sophisticated, they will enable more proactive and tailored health care.”

Diego Bernardo

Principal

Smart implants will play an increasingly significant role in patient-centered care. Smart implants are advanced medical devices that integrate with the body’s biological systems to monitor, diagnose, or treat various conditions in real time. These implants are equipped with sensors, microprocessors, and wireless communication technologies, enabling them to gather critical health data and adjust their function based on patient needs. For example, smart cardiac implants can regulate heart rhythms or detect arrhythmias, while glucose-monitoring implants continuously track blood sugar levels for diabetics. Neuro-prosthetics and brain-computer interfaces (BCIs) are also part of this revolution, allowing patients to control prosthetic limbs with their minds or even restore motor functions in cases of paralysis. These devices offer continuous, real-time monitoring and treatment, reducing the need for frequent medical interventions and improving the quality of life for patients with chronic conditions. As smart implants become more sophisticated, they will enable more proactive and tailored health care.

AI-powered N-of-one studies will expand, providing the computational power and advanced analytics needed to make personalized medicine feasible. AI and machine learning are key enablers of N-of-one research. In N-of-one studies, a single patient’s unique data—such as genetic information, lifestyle habits, and continuous health monitoring—is collected and analyzed in real time. AI and machine learning algorithms can process this vast array of individual data, identifying patterns, predicting health outcomes, and optimizing treatments specific to the patient. These tools allow continuous learning from a patient’s evolving responses to interventions, enabling dynamic adjustments to therapies based on real-time feedback. This personalized, data-driven approach can lead to more effective treatments and improved patient outcomes, making AI and machine learning critical to the future of precision medicine.

Vocal biomarkers that detect early signs of disease have the potential to speed interventions and improve patient outcomes. Vocal biomarkers represent a cutting-edge trend in digital health, where subtle changes in voice patterns are analyzed to diagnose and monitor various health conditions. By using artificial intelligence and machine learning algorithms, vocal biomarkers can detect early signs of diseases such as Parkinson's, Alzheimer's, respiratory infections, and even mental health disorders like depression and anxiety. These tools analyze factors such as tone, pitch, cadence, and even micro-tremors in the voice, offering a non-invasive, scalable method for continuous health monitoring. Vocal biomarkers hold particular promise in telemedicine, where remote assessment is critical. As the technology advances, voice analysis could become a routine tool in both preventive care and chronic disease management, offering early intervention possibilities and improving patient outcomes through real-time data collection. This approach also aligns with the growing trend of passive health monitoring using everyday interactions.

As AI matures, it is rapidly expanding possibilities for patients, providers, and health care organizations.

“Resilient manufacturing processes will look to AI as a strategic enabler, helping pharma manufacturers meet growing demand with improved accuracy.”

Satty Chandrashekhar

Managing Director and Partner

Drug discovery and development continue to accelerate. Forward-thinking pharma and biotech organizations will continue to reshape their R&D agendas, leveraging customized language models to improve understanding of disease biology and accelerating processes to identify promising compounds. Models, both commercial and open, already present the potential to analyze vast biomedical data sets to suggest novel molecular structures or predict drug interactions. Combined with causal modeling approaches, the ability to identify clues previously undiscovered or underrepresented in clinical data will continue to evolve. And in 2025, this trend will further shorten discovery cycles and reveal more promising candidates to test in clinical settings. Clinical development will also continue to accelerate. By using AI to improve data quality, better understand data lineage, and enable evolved uses of operational and patient data to find the right sites and more precise populations for clinical studies, the industry will force a reckoning with operational data readiness.

Quality control comes of age in pharma manufacturing. Generative AI (GenAI) has started to play a transformative role in ensuring higher standards of quality control in the manufacturing of pharmaceutical and medical device products. Leveraging this capability to enhance the detection and mitigation of deviations in manufacturing processes, AI will help organizations address quality issues in more standard ways across manufacturing facilities, many of them global, by analyzing vast operational data streams from production environments. This approach to quality control will allow manufacturers to adjust processes, reduce waste, improve yield, and increase product quality. An issue-resolution GenAI solution trained with historical data, for example, has the potential to help organizations assess the effects of minor changes on product outcomes, enabling companies to reimagine processes without extensive and often manual trial-and-error tests. Enhancing safety while staying compliant with regulations is critical to this effort—and at-scale can accelerate the speed at which new treatments reach the market. Resilient manufacturing processes will look to AI as a strategic enabler, helping pharma manufacturers meet growing demand with improved accuracy and lower production costs.

Pharma commercial enterprises will reimagine how they make data-driven decisions. Many pharma companies continue to transform how they generate insights and make strategic decisions. With the proliferation of real-world data from sources like electronic health records, patient registries, data related to social determinants of health, and other non-traditional sources, companies are harnessing AI to derive actionable insights at unprecedented speed and scale. AI platforms will synthesize complex data sets into clear insights that inform everything from market access strategies to patient engagement and salesforce optimization. Commercial teams will increasingly rely on predictive models to forecast trends, identify emerging therapeutic needs, and optimize pricing strategies in business-time. These models will allow mature companies to pivot quickly based on evolving global dynamics, competitor actions, and regulatory shifts. This will lead to new operating models and new ways of working to harness this data across the "extended enterprise" with speed and precision—to move beyond traditional silos and integrate information to drive more cohesive decisions in the organization and anticipate and respond to trends more effectively.

The growing influence of GenAI on every aspect of health care—from personalized care to automated workflows—is a key theme.

“Consumer behavior patterns will likely force established online health information gateways to offer their own bespoke AI tools.”

Nick Cristea

Vice President,
Experience Design

Consumers will increasingly use AI chatbots for health questions, and a growing number of people will see them as trusted resources. A KFF poll from August 2024 found that one in six adults say they use AI chatbots at least once a month to find health information or advice, rising to 25% for adults under 30 years old. As the technology improves, these consumer behavior patterns will likely force established online health information gateways to offer their own bespoke AI tools or risk losing web traffic. Once providers move past their risk-averse strategies, they will be able to start realizing significant operational efficiencies and competitive advantage by leveraging their “clinical expert” brands to attract patients to their AI services, while also reducing the burden on humans who staff the 24/7 triaging capabilities that they offer. Deploying these tools first as co-pilots for their “nurse line” staff provides an early stepping stone for building and testing their capabilities and ensuring that they do not get bypassed in the future as patients seek greater access to immediate answers and strategies to relieve their symptoms. The providers who architect these solutions most effectively will be able to realize a host of downstream opportunities in attracting patients, collecting data, and increasing effectiveness in triaging and routing patients to the most appropriate sources of care.

At-home diagnostic solutions will become a more ingrained part of how patients and providers interact. These tools should help simplify logistics for patients, especially those in more rural settings. One example is TytoCare, a portable medical exam kit with an app that sends information to providers. The company’s recent announcement of a new integration with Epic’s MyChart, in collaboration with the University of Miami Health System, will leverage TytoCare for medical exams and enable asynchronous workflows for remote patient monitoring and primary care. With big hardware companies continuing to invest in consumer health care solutions, there will be an increasing expectation of preventive care benefits rather than simply vitals monitoring, and more formalized partnerships will emerge. As clinical researchers discover new ways of detecting the early onset of disease through measurable biomarkers, patients with a history of chronic conditions will be able to sign up with remote monitoring programs, which in turn will feed more data back to the research teams. And, with the collation of richer personalized health data, providers will gain a deeper understanding of how best to automate solutions for patients or escalate to the right people at the right time, improving their operational efficiency while also decreasing time to treatment.

“By interpreting and synthesizing unstructured clinical data into actionable insights, GenAI will streamline workflows and improve efficiency.”

Andre Heeg

Managing Director and Partner

The combination of AI, genomics, and wearable tech is paving the way for highly personalized treatments. Digital therapeutics will evolve to offer precision care based on general population data and individual genetics, lifestyle, and real-time health data. For instance, we’ll see more apps and platforms tailored to individuals managing chronic diseases that can predict flare-ups and suggest real-time interventions based on continuous health monitoring.

AI-powered decision-making tools will become mainstream, improving diagnostics, treatment plans, and patient outcomes. In particular, GenAI will give physicians near-instant access to research insights, treatment guidelines, and real-world evidence, allowing for more informed decisions. This will significantly reduce diagnostic errors and speed up patient care delivery.

GenAI will revolutionize health administration by automating the creation and updating of medical records, reducing physicians’ time spent doing paperwork. By interpreting and synthesizing unstructured clinical data into actionable insights, GenAI will streamline workflows and improve efficiency. This will free up more time for health care providers to focus on patient care while ensuring that health records are more accurate and comprehensive.

“Medicine 3.0 prioritizes healthspan driven by improved prevention, personalization, and participation.”

Iana Kouris

Managing Director

The distinction between longevity and healthspan is becoming more and more important. According to *The Longevity Imperative*, by Andrew Scott of the London Business School, “In the UK in 1965, the most common age of death was in the first year of life. Today the most common age to die is 87 years old.” This sounds like great news. However, a longer life doesn’t always mean a healthy life. While the global life expectancy has continued to rise (from 66.8 years in 2000 to 72.5 years in 2020) and the healthy life expectancy (HALE or healthspan) has also increased (from 58.1 in 2000 to 62.8 in 2020), the gap between life span and healthspan has actually increased by 1 year during that time (from 8.7 to 9.7 years). Hence, on average, we spend one more year with disability/in poor health than we used to, according to the World Health Organization. This results in additional medical costs, challenges for our insurance systems, and increased years spent with suffering/discomfort.

The concept of Medicine 3.0 and shifting to proactive approaches in health care is gaining momentum. While Medicine 2.0 focuses mainly on extending lifespan and disease treatment, Medicine 3.0 expands this and prioritizes healthspan driven by improved prevention, personalization, and participation, as noted in *Outlive* by Peter Attia. This drives demand for regular heath checks, popularizes prevention approaches, and leads to the emergence of new businesses in these areas.

Generative AI in health care is driving better access, personalization, and quality. In particular, GenAI can support health care in the area of Medicine 3.0. For instance, it can drive personalization through chatbots, virtual assistance, and more precise analysis of health-related data. New GenAI models focused on medical applications will be developed that can, for example, interpret data from medical imaging, lab results, and electronic health records to produce written or spoken recommendations.

“If 2023 was about GenAI experimentation and 2024 was about point solutions, 2025 will be about value delivery through end-to-end transformation.”

Julius Neiser

Managing Director and Partner

More than a third of ongoing GenAI programs will fail to deliver value in 2025, and some health care players will draw the wrong conclusions from that. Many GenAI solutions are delivering true impact by, for example, reducing medical/regulatory writing effort by 50% and shaving valuable months off drug launch timelines. But more than a third of programs fail. The takeaway from these failures should not be to reduce funding, but rather to build on the lessons learned: obsess about business outcomes, rigorously track key performance indicators, and concentrate on the important people aspect of GenAI transformations. The failure of some programs will ultimately pave the way for more sustainable and impactful transformations, driving a sharper focus on integrating GenAI into existing health care workflows.

We will see an evolution from health care GenAI point solutions to agentic end-to-end process transformations. If 2023 was about GenAI experimentation and 2024 was about point solutions, 2025 will be about value delivery through end-to-end transformation. Instead of isolated AI tools focused on specific tasks like physician note taking or scheduling, we will witness integrated systems that automate entire workflows, for example, from patient intake to treatment plans. These intelligent agents will coordinate across departments, learning from each interaction to improve efficiency and outcomes. For example, in pharma, key processes that will be transformed with GenAI include clinical trials, regulatory submissions, medical legal regulatory review, and omnichannel engagement.

Unstructured health care data will become the new structured health care data. Advances in AI-driven data processing will allow systems to analyze and organize vast amounts of medical records, clinical notes, and physician/patient interactions previously considered too disordered to leverage. This shift will enable health care professionals, pharma companies, payers, and providers to extract actionable insights from a much larger data set, improving patient care and outcomes. This transformation will enhance interoperability across different systems, facilitating seamless collaboration between providers and empowering more personalized and precise medical treatments for patients.

“GenAI chatbots will reshape the cost profile of contact centers and drive significant efficiencies.”

Etugo Nwokah

Managing Director and Partner

GenAI chatbots will define a new table-stakes for payers and providers to deliver an omni-channel customer experience that patients and members will expect. These capabilities will reshape the cost profile of contact centers and drive significant efficiencies. Health plans will see a significant improvement in the adoption of self-service capabilities by their members that will drive major cost efficiencies, which will be reinvested to support ambitious growth agendas and M&A.

Organizational models that focus on digital products and services will further evolve in hospital systems and payer organizations of all sizes. As Chief Digital Officer or Chief Product Officer roles become more prevalent in these organizations, principles around building technology with measurable outcomes and customer-centric experiences will overcome historical challenges of IT functions becoming “digital feature factories” building a lot of capabilities that no one uses.

Further innovation around clinical specialties will drive increased value-based payment arrangements and end-to-end management for specific diseases. Specialties such as oncology, orthopedics, and behavioral health will continue to see higher spending by payers, whose ability to influence and measure outcomes, well-defined populations, and episodes of care will be crucial to leveraging new technology enablers such as large language models (LLMs).

“Forward-thinking health care companies will focus on driving AI adoption internally to accelerate scaling.”

Sid Thekkepat

Managing Director and Partner

Many health care AI use cases will be slow to scale and drive value, leading to increased pressure on IT and change management teams. Winning health care companies will experiment with new approaches to drive adoption. By fostering a culture of experimentation, collaborating with frontline staff, and prioritizing user-friendly AI solutions, these organizations can enhance adoption rates and realize AI’s full potential. Health care companies that thrive will be those that treat adoption as a critical aspect of AI implementation, leveraging iterative learning and adaptive frameworks to drive sustainable value.

As valuations drop, consolidation and M&A activity will intensify in the health tech space, leading to the emergence of more scaled, sustainable platforms. Larger strategic players will seize this opportunity to strengthen their positions by acquiring or merging with innovative but financially constrained startups. Private equity and venture capital firms, facing market uncertainty, may hold back for now, allowing established companies to lead the charge with bold and creative deals.

Sleep monitoring is emerging as a mainstream biomarker that consumers are increasingly prioritizing in their health-tracking routines; providers and life sciences will pay more attention. As awareness grows around the critical role sleep plays in overall well-being—impacting everything from mental health to chronic disease prevention—health tech companies are responding by focusing on developing more advanced, accurate sleep-tracking tools. This trend is driving innovations in wearables, apps, and even non-invasive monitoring devices designed to provide deeper insights into sleep patterns, quality, and its correlation with other health metrics. Expect sleep tracking to become a cornerstone of personalized health solutions as tech continues to refine its capabilities.

“The focus on tech-enabled mental health will continue to grow, notably by further integrating mental health services into primary care.”

Gunnar Trommer

Managing Director and Partner

More providers/hospitals will develop their own workflow efficiency solutions leveraging GenAI, mainly aiming at reducing the administrative burden on clinicians and improving workflows. As long as providers develop tech-enabled solutions for their own patients and clinicians to enhance clinical and administrative workflows, they do not have to seek regulatory approval for the tools. Because the latest GenAI technologies make developing these tools much easier, providers are more inclined to create or tailor their own solutions. Examples include speech-to-text capture of patient/physician conversations and automatically drafting discharge notes.

Post-pandemic mental health concerns and societal pressures continue to drive more demand for scalable, tech-enabled mental health services, including services that address cognitive decline. As a result, the focus on mental health will continue to grow, notably by further integrating mental health services into primary care. Digital mental health tools such as apps and AI-driven chatbots will make it much easier to access therapy and support. There will still be a “human in the loop” aided by technology to counter clinician shortages and make mental health services more accessible and more cost-scalable.

Products and solutions that are enabled by machine learning—and soon, GenAI—will see accelerated adoption in diagnosing diseases, analyzing medical imaging, and predicting patient outcomes. AI-powered tools will help clinicians make faster and more accurate decisions—especially in areas like oncology, radiology, cardiology, and pathology—through a combination of pattern recognition (using traditional AI/machine learning) and content generation (GenAI).

“There is a huge opportunity to redesign traditional ‘hardware’ used for women’s health, with the female experience at the center.”

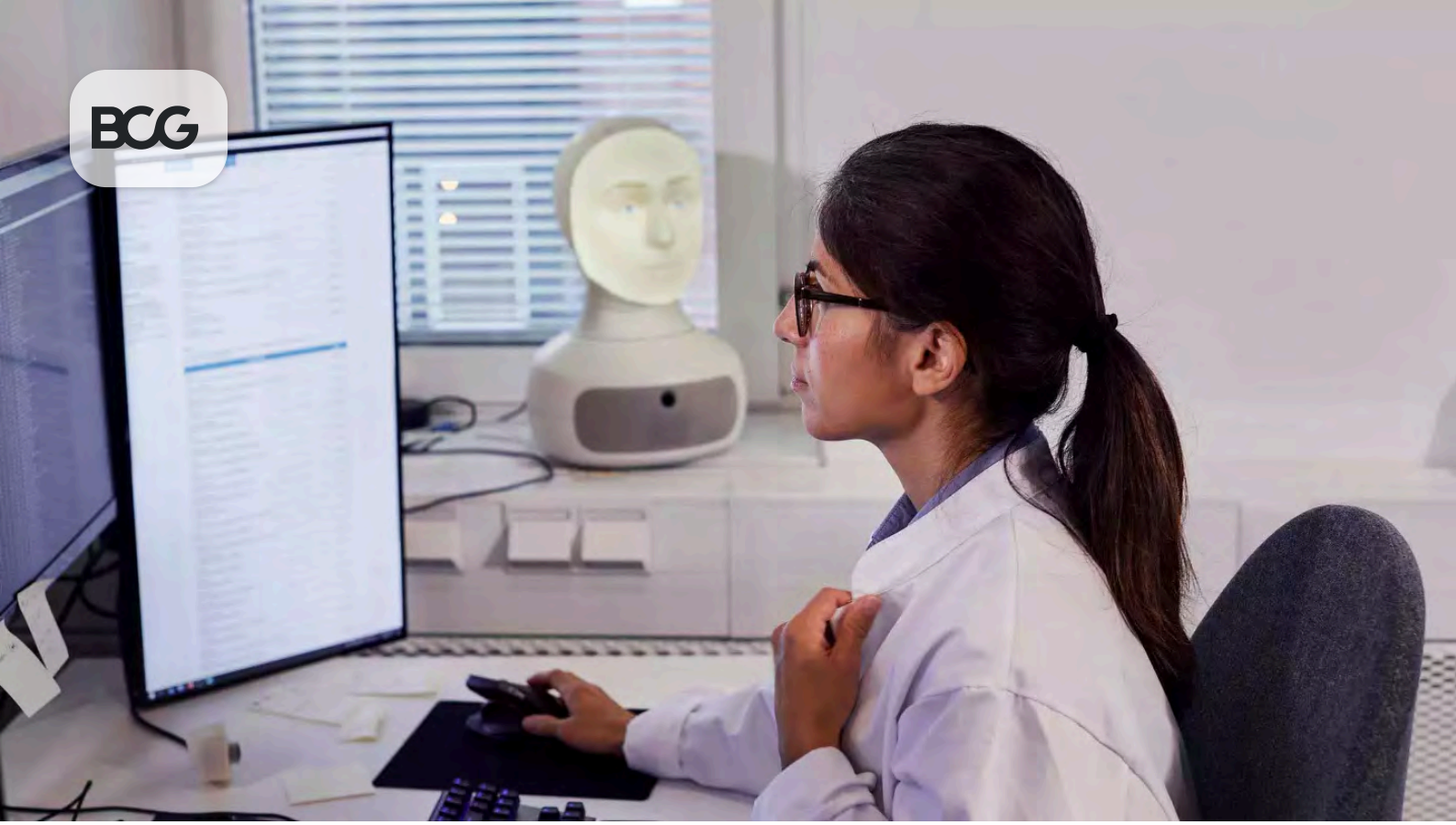
Alice Wilson

Principal

Femtech partnerships and ecosystems start to mature. In 2024, we saw some promising partnerships between femtechs and health and wellness brands. The smart ring developer OURA has really been a trailblazer, developing partnerships with a range of femtechs such as Natural Cycles and Clue that combine biometric data with menstrual cycle tracking. In 2025, we predict that we will see more mature models of these partnerships emerge, based on clear data exchanges. It is only by creating interoperable ecosystems that can pool critical data on women’s health that we will be able to gather insights to drive improved health outcomes.

There will be a leap forward in femtech “hardware” enabled by software. The speculum has remained the same design for 150 years, and the pain associated with its use is often cited as the number one reason that women miss vital cervical cancer screenings. There is a huge opportunity to redesign traditional “hardware” used for women’s health, with the female experience at the center, combining it with software to accelerate diagnoses. We’ve seen glimmers of innovation: at-home cervical smear tests and smart tampons are starting to appear. But we believe there will be further breakthroughs in 2025.

The femtech sector, particularly in the West, is witnessing saturation in the B2C market. Intense competition is making it difficult for many digital-first offerings to commercialize successfully. In response, the industry is shifting toward B2B models, partnering with employers, health care providers, and insurers to embed women’s health solutions into broader systems. While this shift demands femtechs to robustly demonstrate the return on investment to their new partners, those adopting value-based care models with evidence of improved outcomes and cost savings are best positioned to succeed in this evolving landscape.



APPENDIX B

GENERATIVE AI

Generative AI Will Transform Health Care Sooner Than You Think

By Matthew Huddle, Josh Kellar, Krishna Srikumar, Krishna Deepak, and Daniel Martines

ARTICLE JUNE 22, 2023 12 MIN READ

Generative AI has quickly become a major factor in a number of industries, including health care. It has the potential to transform the sector, but executives must understand how to use the technology in order to capitalize on its potential while avoiding the risks inherent in applying it to patient care. We analyzed numerous use cases across health segments, looking at both solutions already on the market and those likely to arrive soon. Our goal was to demystify generative AI and help leadership teams make sense of the options.

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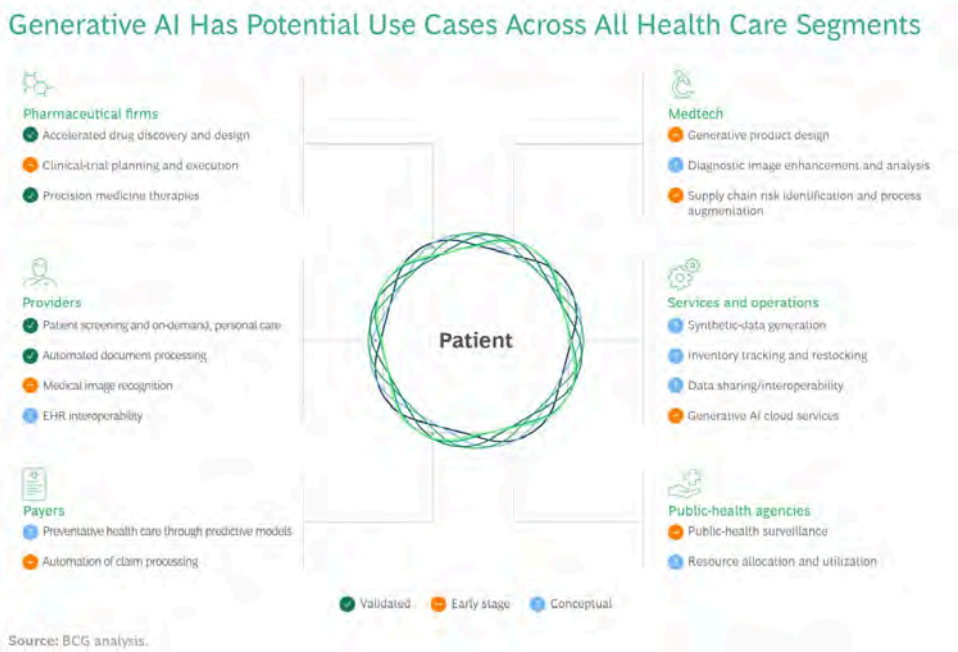
Generative AI algorithms can analyze large volumes of medical data and create entirely new content. The technology can improve the quality of care, make it more accessible and affordable, reduce inequities in research and care delivery, and help companies unlock value in new ways . Moreover, generative AI overcomes some of the previous hurdles to AI adoption in health care. It requires less data, is more adaptable to unfamiliar situations, and can interface better with clinical staff. These features make generative AI more broadly applicable and transferable to different health care tasks.

Emerging Use Cases in Key Segments

Generative AI has potential use cases across the health care industry . (See the exhibit.) For each segment—providers, pharmaceutical firms, payers, medtech, services and operations, and public-health agencies—we grouped the options into three categories: validated solutions already on the market, early-stage or conceptual use cases, and potential future use cases not yet in development.



Providers

Several generative AI providers are developing solutions—from diagnosis to care provision to patient monitoring—to help providers improve clinical outcomes. Others are working to improve resource utilization by both clinical and administrative staff.

Validated Products in Use Today. Paige.AI, a digital pathology company, is integrating generative AI into its products to improve the accuracy and efficiency of prostate cancer detection. It was the first company to receive FDA approval for AI use in digital pathology and is looking to integrate the resulting information into patient electronic health records along with other clinical data.

On the administrative front, Doximity, Abridge, and DeepScribe are exploring applications that automate processes such as documentation, claims handling, preauthorization and appeals, patient onboarding, and scheduling. DeepScribe, which offers AI scribing services, has been able to decrease the amount of time providers spend on administrative tasks by three hours each day, and Abridge's ambient AI scribing products are now in use at more than 140 provider locations in the University of Kansas Health System.

Early-Stage/Conceptual Use Cases. To aid caregivers, some providers are developing digital solutions that patients interact with directly. For example, Babylon Health has created a digital health service that uses generative AI to understand patients' evolving risk profile, helping providers offer more personalized care at lower cost. A generative AI-powered counseling chatbot available on demand from Serena delivers accessible and affordable mental health care.

Potential Future Use Cases. In the future, generative AI could support real-time patient monitoring, along with data analysis to generate personalized insights that encourage healthy behaviors or lead to timely interventions before medical conditions worsen. Generative AI could also make imaging solutions more accurate and transferable across different practice areas. Finally, the technology's adaptability and interactivity could encourage preventive care, wellness, and healthy behaviors via personalized nudges on mobile apps, wearables, and monitoring devices.

Pharmaceutical Firms

Generative AI is accelerating drug discovery, improving clinical-trial planning and execution, and leading to more precision medicine therapies.

Validated Products in Use Today. Generative AI allowed Insilico Medicine to go from novel-target discovery to preclinical candidate in just 18 months, spending only $2.6 million. The company's idiopathic pulmonary fibrosis drug recently received the agency's Orphan Drug Designation after completing the preclinical phase in 30 months, much faster than average for a new treatment.

"Generative AI is accelerating drug discovery, improving clinical-trial planning and execution, and leading to more precision medicine therapies.

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The biotech company Exscientia is using generative AI to analyze patient tissue and employ functional precision oncology to improve patient outcomes. NVIDIA is offering a set of generative AI cloud services that enable customization of AI foundation models to accelerate drug discovery and research in genomics, chemistry, biology, and molecular dynamics. The services provide pretrained models and enable researchers to fine-tune generative AI applications on their own proprietary data. The offering has been adopted by drug discovery startups such as Evozyne and Insilico Medicine, as well as by incumbents such as Amgen.

Early-Stage/Conceptual Use Cases. Several biotech firms are at an earlier stage of generative AI exploration:

- Ordaōs is developing a class of miniproteins with enhanced biological and therapeutic attributes to treat uniquely challenging rare cancers that primarily afflict ethnic minorities..
- Absci is using deep-learning AI and synthetic biology to design new antibodies against cancers and immune diseases.
- Profluent created a machine-learning model able to generate new protein sequences with specific functions, an innovative tool that can help improve access to affordable treatments.

Beyond drug discovery, generative AI could accelerate and improve clinical trials and precision medicine therapies. For example, digital modeling of clinical trials—including synthetic control groups—has recently been validated. Similarly, a tool developed by Synthesized can help researchers expand existing drugs beyond their initial use to treat other diseases, which could make medicines more accessible.

Potential Future Use Cases. Looking ahead, the use of generative AI at the preclinical and clinical stage could accelerate access to therapeutics, even for rare conditions whose treatment development has been difficult or economically prohibitive. The technology may be also used in the analysis of patient data to identify subgroups likely to respond to specific treatments or to personalize drugs to the unique needs of individual patients.

Payers

Payers are starting to leverage generative AI to reduce costs and improve risk management and member engagement, with the overall goal of offering higher-quality coverage at less cost to consumers.

Validated Products in Use Today. DigitalOwl is automating much of the underwriting and claims management process, reducing operating expenses and turnaround times to boost affordability.

Early-Stage/Conceptual Use Cases. Companies like ConcertAI are developing predictive models to identify and proactively manage high-risk segments based on patient medical history and demographic and social determinants of health.

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> Payers are starting to leverage generative AI to reduce costs and improve risk management and member engagement.

Potential Future Use Cases. Generative AI could support applications like conversational AI to deliver personalized messages based on member health needs and preferences.

Medtech

Generative AI could help companies create more personalized and patient-centered devices—incorporating software that allows for preventive maintenance and repairs, for example.

Early-Stage/Conceptual Use Cases. The UK 's National Centre for Additive Manufacturing is applying generative AI to optimize the design of medical devices such as prosthetics and implants, tailoring them to the needs of individual patients. And medtech company Implicity is using the technology to incorporate remote monitoring in pacemakers and implantable defibrillators.

In brain health, DiagnaMed recently announced the development of a platform leveraging generative AI to analyze electroencephalography signals in order to predict and monitor brain aging and provide insights and tools in the diagnosis, prevention, or mitigation of cognitive decline in patients with mental health and neurodegenerative disorders.

Activ Surgical, a digital-surgery pioneer, recently announced completion of its first AI-enabled case, which provides enhanced visualization and real-time, on-demand surgical insights inside the operating room.

Potential Future Use Cases. Future applications could enable companies to collect and analyze data via remote-monitoring systems, leading to more effective patient interventions. Quality control applications could predict when devices and equipment may need repairs, allowing caregivers to schedule maintenance and reduce downtime.

Services and Operations

In health care services, generative AI can be particularly useful in data analytics and software optimization. Because it is more flexible than earlier generations of AI and can accommodate different data modes and formats—and even generate synthetic data to complement insufficient data sets—generative AI can improve the interoperability of existing applications, including health and laboratory information management systems.

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Early-Stage/Conceptual Use Cases. Syntegra and Google's EHR-Safe are creating machine-learning models that generate synthetic health care data sets, enabling fast and easy access to flexible longitudinal data that (with adequate privacy protections) can help public-health programs make more informed decisions.

> "In health care services, generative AI can be particularly useful in data analytics and software optimization.

Potential Future Use Cases. Future applications could help health systems in areas such as inventory tracking and restocking, cold-chain logistics, data sharing, and HR functions (including recruitment and training). Additionally, generative AI could help personalize and automate corporate functions, with potential use cases such as generative AI-enabled office applications, auto-generated knowledge management, and human-machine interaction assistance.

Public-Health Agencies

Public-health agencies, other health organizations, and government ministries could leverage generative AI to improve resource planning and allocation, anticipate public-health needs and interventions, and execute programs more effectively.

Early-Stage/Conceptual Use Cases. BioNTech recently acquired InstaDeep in order to develop an early-warning system for new COVID-19 variants. Structural modeling of the SARS-CoV-2 protein combined with InstaDeep's generative AI capabilities allows the system to proactively alert researchers, vaccine developers, health authorities, and policymakers.

Potential Future Use Cases. Going forward, generative AI-powered tools could be used to monitor public health and allocate resources. In the US, Medicaid could potentially leverage the technology to better manage allocations based on health data and forecasted need. The FDA could use it when reviewing the safety and efficacy of drugs, and generative AI could help public-health groups like Doctors Without Borders predict outbreaks and mobilize resources to minimize impact.

Managing Potential Risks

Although generative AI technology is promising, some near-term caution is warranted. There are several inherent risks that providers must address before broad adoption in health care can occur.

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- **Biased Outputs.** Generative AI results can reflect inherent biases in the underlying data. In response, generative AI companies need to assign experts to review the data and results and correct for bias through oversampling and other statistical techniques.
- **False Results.** Because the models are still evolving, they can sometimes generate results that are simply wrong (a phenomenon in AI known as hallucinating). Providers will need to make their models more transparent and emphasize the need for human review of outputs.
- **Patient Privacy.** Patient health data is sensitive and needs to be handled with extreme care. Companies with generative AI solutions should clarify data ownership with partners, strengthen cybersecurity, and look beyond existing data to the development of synthetic data.
- **Opaque Results.** A challenge with generative AI is that it operates as a black box, making it unsettling to some users. To build trust and increase adoption rates, health care organizations will need to explain how a given algorithm works and how a specific set of data leads to a prognosis.
- **Misuse or Overreliance.** Patients may rely too heavily on information from generative AI. Hospitals, clinicians, and payers should clarify how specific solutions should be used, with clear messaging that AI-generated insights are recommendations rather than mandates.

How to Get Started

Even as the range of generative AI applications continues to expand, incumbent health organizations can create the right foundation for adoption and implementation.

- **Create an enterprise-wide generative AI strategy.** Explore new offerings and business models with generative AI.
- **Build data systems.** Invest in data management and analysis tools to leverage large and mixed data sets.
- **Invest in capabilities.** Recruit in-house experts to advise, implement, and monitor generative AI opportunities and train key stakeholders in using the technology to make informed decisions.
- **Forge strategic partnerships.** Partner with tech players and consulting firms to bring in timely and best-in-class external assets and expertise.
- **Integrate with the broader ecosystem.** Establish data interoperability, offerings, and systems foundational for generative AI. Ensure close coordination with regulatory agencies to co-develop effective and safe solutions.

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APPENDIX C

DIGITAL GOVERNMENT

How Governments Can Keep Digital Transformations on Time and on Budget

By Ashwin Bhave, Rick Berger, Rob Kurtz, Jon Kalscheuer, Matthew Leybold, and Crosby Kapp

ARTICLE OCTOBER 16, 2024 8 MIN READ

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US government agencies continually struggle to execute digital transformations. Programs frequently run over budget and over schedule. Worse, after spending billions of dollars on multi-year programs, agencies (both civilian and defense) often end up with limited gains at best, and no meaningful change in capabilities. This approach is no longer sustainable. Instead, the government must rethink its approach to digital transformations.

Some public sector organizations have taken steps to improve, such as creating a flexible technology architecture, working in more agile ways, and being more transparent about program status. However, those measures—while important—are becoming table stakes. Governments also need to take a capability-led approach to transformation by focusing on the ultimate goals they are seeking to achieve. Differentiators such as shifting the mindset from project management to overall impact, rethinking contractor and vendor management to enforce accountability for outcomes, and planning for the overall transformation life cycle from the start can all help governments build capabilities more effectively in transformations.

Moreover, governments require a new role to enable those differentiators: an unbiased, third-party navigator that can serve as a trusted advisor and liaise between the government and the prime vendors and system integrators that it hires. In our experience, governments that use a navigator to help implement the differentiators can flip the odds of a transformation dramatically in their favor and capture the full potential value of a transformation. In that way, they can keep large-scale change initiatives within budget and on schedule, emerge with stronger capabilities, and better equip themselves to accomplish their mission.

The Growing Digitization Imperative

The US government has a growing imperative to digitize and modernize. Core systems at many agencies are becoming obsolete, and regulatory requirements are growing. At the same time, new solutions can help organizations dramatically improve their performance through data-driven insights, even as citizens' expectations are growing for a more intuitive customer experience.

When executed well, digitization can lead to dramatic advances in government services in areas such as improving health care patient outcomes and creating a more intuitive and customer-friendly experience for taxpayers and constituents. Internally, digitization can improve capabilities, streamline back-end processing functions, boost efficiency, and make data more accessible to the government leaders and employees who protect and serve our nation.

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That said, digital transformations are large, complex programs that frequently fall short of their objectives. Among both public and private sector initiatives:

- only 35% of large transformations succeed within timeline and budget;
- 98% of large transformations face unforeseen implementation difficulties, leading to scope creep and additional requirements; and
- programs can take five to eight years before they begin generating financial value—a lifetime given how fast digital technology advances.

Based on our experience supporting government transformations over the past ten+ years, BCG has determined that there are several underlying issues that cause large-scale digital transformations in government agencies to go off-track. These include a waterfall-based, tech-first approach, insufficient governance, and an unclear strategy, among others. (See Exhibit 1.)

Exhibit 1 - Government Digital Transformations Experience Several Common Problems

Underlying issue	Signpost of trouble
Unclear strategy	"We're four years into the transformation, and I'm not sure where we're heading or what differences end users will actually see."
A waterfall-based, tech-first approach	"We can't change anything until the new technology is 100% nailed down."
Insufficient governance	"The vendor told us everything's on track, but we have no way to validate or effectively dictate changes in requirements."
Misaligned incentives with vendors	"We have separate contracts to develop, field, sustain, and train on the new system."
Overreliance on commercial off-the-shelf software	"We bought the latest solution, but we haven't changed any of our business or operational processes to use it."

Source: BCG analysis.

Table-Stakes Measures to Improve Digital Transformation

Our experience also points to the following table-stakes measures government organizations can implement to make digital transformations more successful. These measures are not new, but in our experience, many government agencies still don't put them into practice:

- **Build a flexible, extensible technology architecture.** Emphasize flexibility, adaptation, and scalability. A digital and data platform can standardize core transactional systems and data

platforms, and separate data from individual applications or systems.

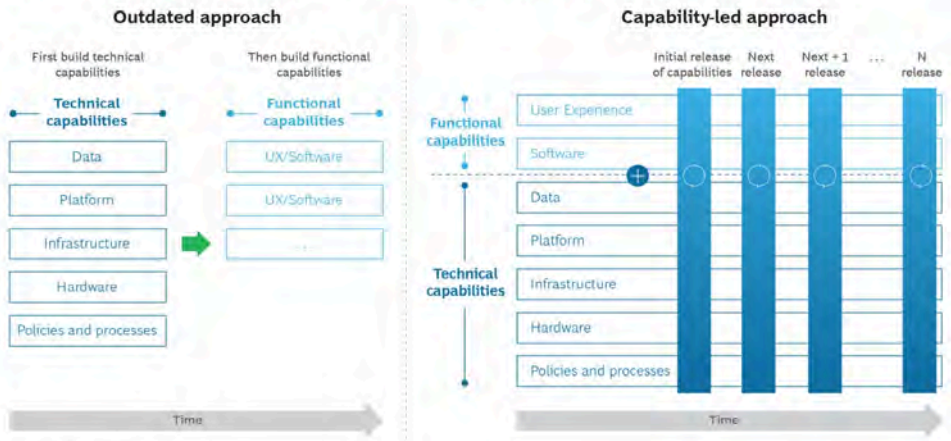
- **Adopt agile ways of working.** Focus on iterative development through sprint cycles and quick value delivery, using DevSecOps and collaboration tooling to enable speed.
- **Make transformation progress more transparent.** Both stakeholders in government agencies and vendors must have access to a baseline of information—both processes and data—to make informed decisions and ensure that vendors are accountable.

At many government organizations, these measures would lead to significant progress. But they're not enough to execute a digital transformation effectively, on time, and within budget.

Putting New Capabilities at the Center of the Effort

In addition to table-stakes measures, government entities should ensure that a transformation achieves its main objective: building new capabilities for an organization. Rather than a linear, gated process, this often means an iterative approach that addresses people, processes, tech, data, and systems simultaneously to build capabilities incrementally over time. (See Exhibit 2.)

Exhibit 2 - A Capability-Led Transformation Builds Functional and Technical Capabilities Simultaneously, Leading to Faster Progress



Source: BCG analysis.

Three differentiators can help government organizations take a capability-led approach to transformation. (See Exhibit 3.)

Exhibit 3 - Key Differentiators to Navigate a Digital Transformation

	From	To
Shift the organizational focus from project management to overall impact	Focus on efficiency and cost optimization	**Focus on outcomes and impact**
	Overemphasis on technology implementation	**Capability-led approach**
	Prime vendor-owned agenda	**Government-owned agenda with third-party support**
Rethink vendor engagement, including contracting	Time- and material-incentivized contracts	**Performance-based contracts**
	Monolithic prime vendor or system integrator	**Best-of-breed multi-vendor ecosystem**
	Vendor lock-in	**Modular open system architecture with govt IP rights**
Consider the entire transformation life cycle—with a particular emphasis on capabilities	Tech platform built first, functional capabilities built second	**Iterative functional capability with tech build**
	Data as an afterthought	**Data governance at the core**
	End-user experience as an afterthought	**End-user experience embedded from the beginning**

Source: BCG analysis.

Shifting the organizational focus from project management to overall impact. Putting the government in control of an overall program ensures that big-picture objectives and capabilities remain the primary focus. Rather than transforming simply to reduce costs and increase efficiency, agencies can focus on broader and more impactful goals, treating the transformation as an opportunity to rethink what is possible for an organization.

This approach, grounded in clear, centrally agreed-upon metrics for the transformation that align with strategic goals and desired outcomes, extends to governance as well. Metrics should be both holistic enough to reflect overall performance and limited to a discrete number that can be maintained in near real time. Collaboration across departments and vendors should be encouraged for better decision-making.

Rethinking vendor engagement, including contracting. Traditional government contracts can often create the wrong incentives. Instead, agencies should shift to contracts that both reward performance and outcomes and are tied to the achievement of specific program milestones.

Rather than relying entirely on a system integrator or prime contractor, government organizations can create a best-of-breed, multi-vendor ecosystem. This approach allows an organization to optimize skill sets by selecting niche vendors to support the system integrator or prime vendor in implementing technology. To be clear, this approach requires the government to have stronger capabilities in program and vendor management.

Vendor incentives should support the creation of a modular, open system architecture that is flexible enough for subsequent add-ons. Equally important, the government should retain rights to the code and data, along with control over the technology ecosystem, to avoid getting locked into a specific vendor. Prioritizing data governance from the start supports the integration of solutions upon delivery and into sustainment.

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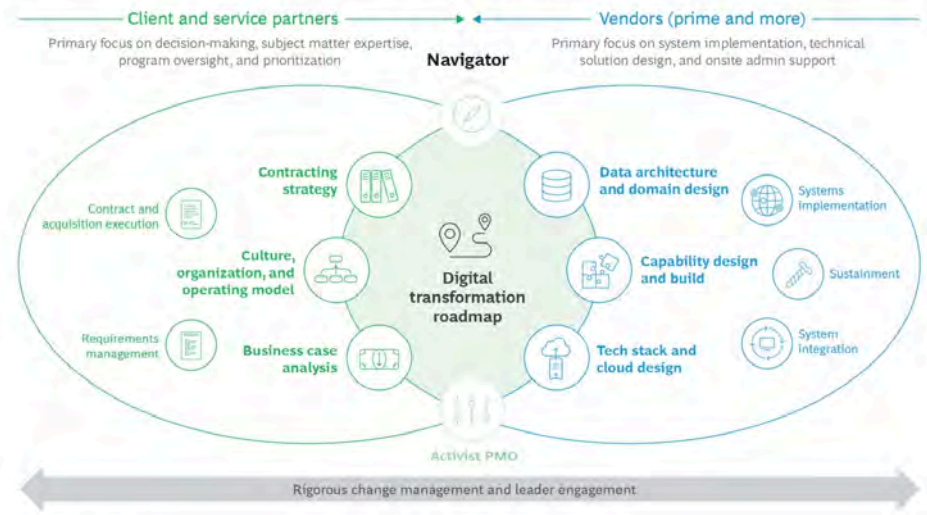
Consider the entire transformation life cycle—with a particular emphasis on capabilities. Rather than waiting for a complete solution to be in place, government customers should build capabilities iteratively and in parallel with the development of a tech platform—an approach that leads to continuous delivery of value. Similarly, governments should embed the end-user experience into a transformation from the start, incorporating feedback from real-world users during the design and testing phase.

Leading organizations plan for the full life cycle of a transformation, from development through sustainment, to ensure long-term viability. They also shift from isolated assessments of individual use cases to a holistic, portfolio-driven strategy that factors in the scalability of a platform and underling capabilities. And top performers consider programmatic requirements—including funding—that are needed to scale pilots and proofs of concept through to full implementation. For example, flexible contracts and funding models support on-demand fielding of new capabilities, updates, and enhancements to systems.

The Power of a Navigator

In many cases, we've seen the government leverage an objective third party—a navigator—to enable the differentiators above and help implement a digital transformation. A navigator isn't an individual, but rather a team of product managers, IT architects, program-specific subject matter experts, and user-experience designers. Collectively, this group serves as a trusted advisor to the government and as a liaison with vendors, bringing the right stakeholders to the right conversations at the right time to ensure better outcomes. (See Exhibit 4.)

Exhibit 4 - A Navigator Liaises Between the Client and Vendor to Develop the Tech Strategy and Pressure-Test All Recommendations



Source: BCG analysis.

This is a flexible approach that can benefit government transformations at all stages and objectives. For new programs, a navigator can define a business case, shape the roadmap, and support vendor negotiations to get the best deal. For programs already underway, a navigator can get timelines back on track, support executional certainty, and make changes sustainable over time. (See “Critical Questions for Government Leaders.”)

CRITICAL QUESTIONS FOR GOVERNMENT LEADERS

Government leaders considering—or undergoing—a digital transformation should assess their needs by asking themselves the following questions:

- Where are we in our transformation journey? Is this a new system, or are we upgrading an existing capability or platform?
- If we’ve already started, are we on schedule and on budget?
- Do we own the transformation agenda?
- Is the program designed to meet the evolving capability needs of stakeholders?
- What is our current spend and ROI to date?
- Where do we see inefficiencies, areas for improvement, and gaps in capability or communication?
- Are our vendors meeting performance expectations as laid out in service-level agreements or contracts?
- Could we benefit from expanding our vendor base?
- Are we attuned to all potential risks from the transformation?

Once you have considered these questions, a third-party navigator can help you implement the differentiators above to ensure that the transformation meets the needs of your organization.

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Specific responsibilities of the navigator include the following:

Creating the business case, strategy, and roadmap

- Identifying and championing the government's highest-value priorities
- Managing stakeholders and serving as a thought partner to leadership teams

Developing the technology platform and data governance

- Determining technology architecture, developing requirements, deciding on key design elements, and prioritizing use cases
- Pressure-testing product vendor and system integrator recommendations against business objectives and external best practices

Overseeing governance and execution

- Enabling and accelerating the standup of governance bodies, transformation roadmaps, and other deliverables
- Creating transparency, ownership, and alignment across all stakeholders to deliver government's mission
- Providing strategic guidance including industry, architecture design, and change management expertise
- Driving design and execution of solutions to advance broader transformation goals without impacting any ongoing enterprise initiatives or strategic priorities

In our experience, combining the differentiators discussed above with a neutral, third-party navigator can deliver a significant ROI on government digital transformations, including the following:

- 40% faster path to value from new capabilities
- 35% reduction of project scope
- 70% increased degree of standardization
- 35% higher transformation success rate

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- 30- to 60-fold return from investing 5% of program budget in de-risking

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Governments face a growing imperative for digital transformation, but these programs are expensive, complex, and time-consuming. Without the right approach, governments can spend millions of dollars and end up with new technology that doesn't meet their needs, alienates users, and leaves an organization further behind than when it started.

Conversely, governments that take a proactive approach—applying the differentiators we identified and relying on an objective, third-party navigator—can ensure that their digital transformation stays on schedule and on budget and leads to strong capabilities. In that way, they can better position themselves to meet the needs of their stakeholders.

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APPENDIX D

THE MOST INNOVATIVE COMPANIES 2019

THE RISE OF AI, PLATFORMS, AND ECOSYSTEMS

MICHAEL RINGEL

FLORIAN GRASSL

RAMÓN BAEZA

DEREK KENNEDY

MICHAEL SPIRA

JUSTIN MANLY

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INNOVATION IN 2019

Users of Google's email software recently found that Gmail was offering to finish their sentences for them. This new Smart Compose feature relies on Google's expertise in artificial intelligence (AI) and machine learning (ML), along with billions of training examples and the company's cloud-based Tensor Processing technology, to intuit what Gmail users want to say—often faster than they can complete their own thoughts.

In a world where computers can compose notes to your friends, it's hardly surprising that the theme of BCG's 13th annual global innovation survey and report is the rising importance of AI and of platforms that support innovation. This is not an out-of-the-blue development. Our last few reports have highlighted the crucial role of science and technology in innovation, the impact of digital technologies on both digital natives and more traditional industries, and strong innovators' increasing use of various internal and external vehicles to uncover new ideas. This year's survey shows that AI use is rapidly expanding and that many companies are relying more on platforms and their cousin, ecosystems, to support their innovations efforts.

Digital Natives Lead, but Others Are in Fast Pursuit

The companies on BCG's most innovative list for 2019—especially those in the top ten—extensively use AI and platforms. (See Exhibit 1.) Alphabet/Google, which unseats Apple at number one after a 13-year reign, is an "AI first" company, according to its CEO. It has long embraced both platforms (think Android) and ecosystems (consider how Alphabet has worked with others to develop Waymo, its autonomous driving venture). Next in line is Amazon, which, besides using AI in its retail business, has pioneered voice recognition technology (Alexa) and platform-based services (Amazon Web Services). Apple, at number three, helped pioneer voice recognition software (Siri) and provided a premier virtual workplace for app developers with its iOS platform. Fourth-place Microsoft has also evolved from a software company into a provider of AI and platform-based services.

All of the ten highest-ranking companies—and many in the top 50—use AI, platforms, and ecosystems to enable themselves and others to pursue new products, services, and ways of working. For example, McDonald's (number 21) uses an AI algorithm to serve digital menus that continuously change in response to such factors as time of day, day of the week, restaurant traffic, and weather. Philips (number 29) launched an AI platform in 2018 that helps "scientists, software developers, clinicians, and health care providers access advanced analytic capabilities to curate and analyze health care data and offers them tools and technologies to build, main-

EXHIBIT 1 | 2019 Most Innovative Companies

1	Alphabet/Google	11	Boeing	21	McDonald's	31	AXA	41	Dell
2	Amazon	12	BASF	22	Marriott	32	Unilever	42	Walmart
3	Apple	13	T-Mobile[2]	23	Alibaba	33	Salesforce	43	eBay
4	Microsoft	14	Johnson & Johnson	24	Bayer	34	Pfizer	44	HP Inc.
5	Samsung[1]	15	DowDuPont	25	AT&T	35	Stryker	45	ING
6	Netflix	16	Siemens	26	Allianz	36	NTT Docomo	46	BP
7	IBM	17	Cisco Systems	27	BMW	37	Toyota	47	Daimler[4]
8	Facebook	18	LG Electronics	28	SAP	38	Volkswagen[3]	48	Huawei
9	Tesla	19	Vale	29	Philips	39	3M	49	Rio Tinto
10	Adidas	20	JPMorgan Chase	30	Royal Dutch Shell	40	General Motors	50	Hilton

Source: 2018 BCG Global Innovation Survey.
[1]Includes only Samsung Electronics.
[2]Includes only US T-Mobile, not Deutsche Telekom.
[3]Includes Audi and Porsche.
[4]Includes Mercedes-Benz.

tain, deploy, and scale AI-based solutions." BASF (number 12) is collaborating with AI software developer Citrine to build models that identify new materials for capturing carbon dioxide and other greenhouse gases.

Two New Forces

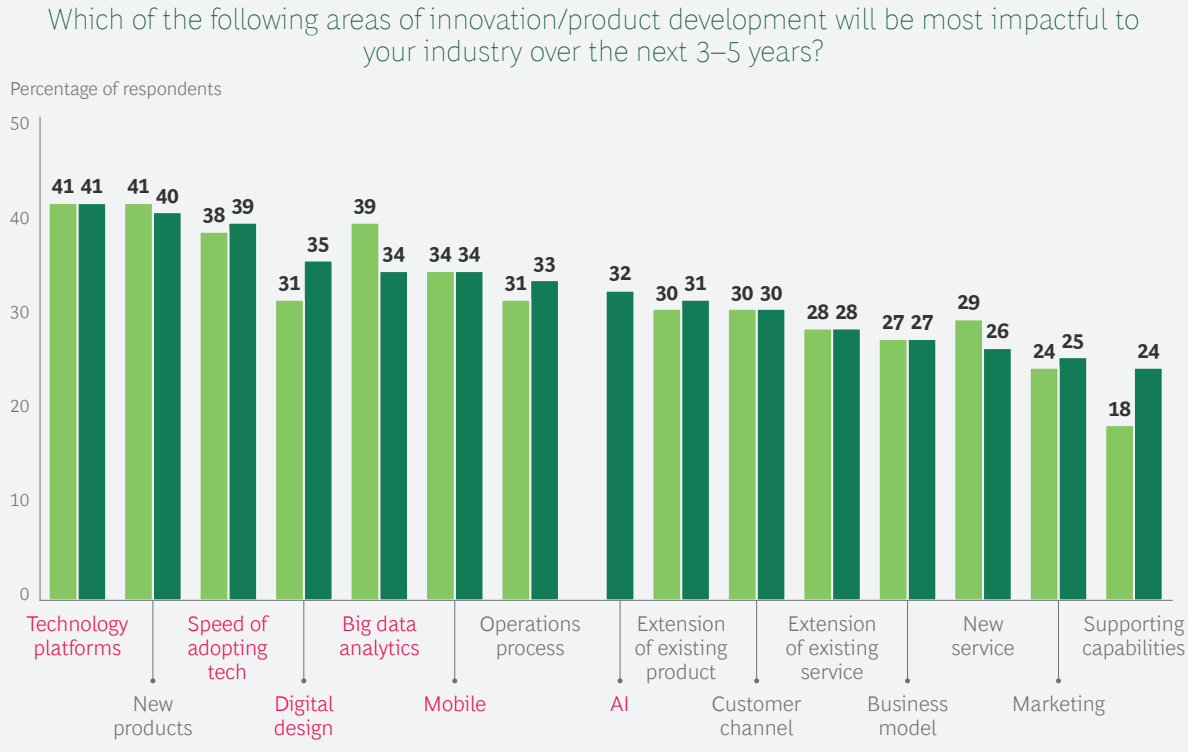
Most companies are at least exploring the use of AI, and strong innovators are seeing positive results. Nine out of ten respondents in our current survey say that their companies are investing in AI, and more than 30% expect AI to have the greatest impact of any innovation area on their industry over the next three to five years. (See Exhibit 2.) Four in ten self-described strong innovators report receiving more than 15% of their sales from AI-enabled products, compared with less than one in ten weak innovators.

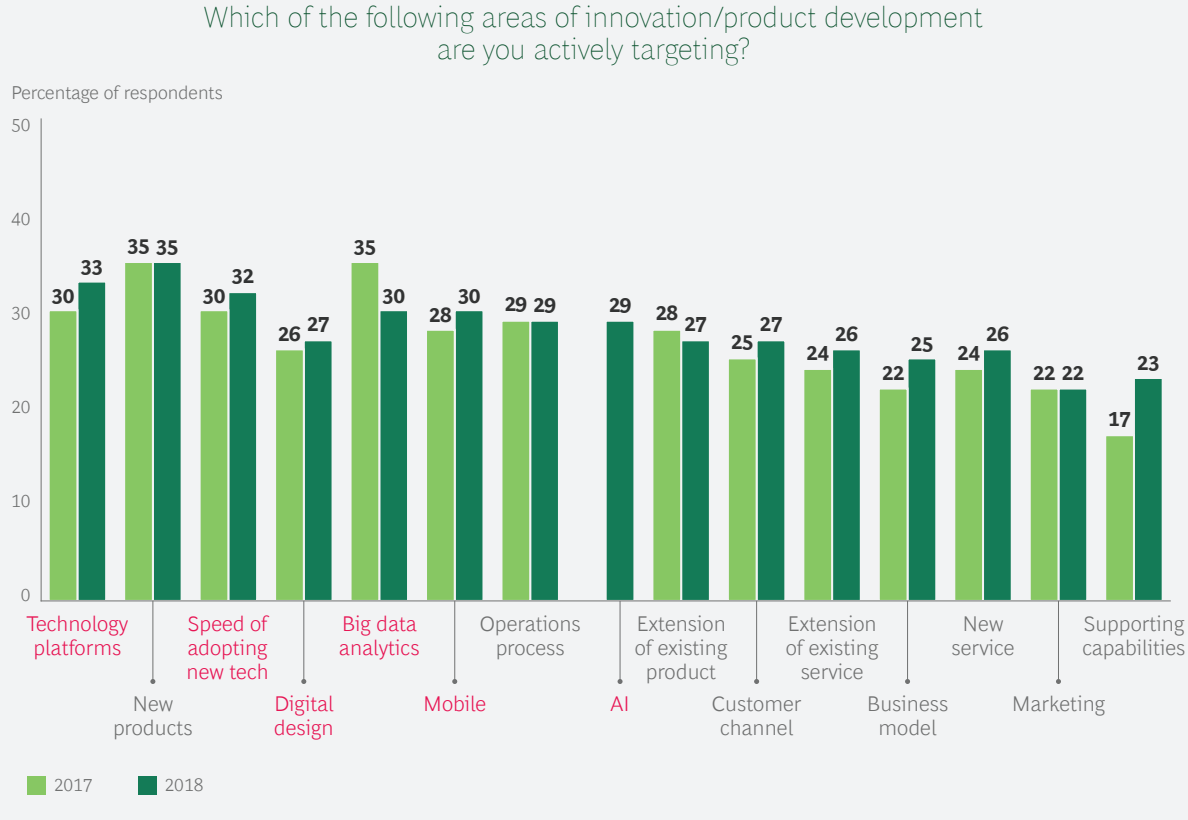
In the next chapter, "AI Powers a New Innovation Machine," we take an in-depth look at the widening gap in where and how AI is affecting innovation.

Platforms and ecosystems serve multiple functions, including facilitating (and sometimes profiting from) the innovation of others, expanding reach and collaboration, and enabling new multiparty solutions and offerings. Again, strong innovators are more likely than weak ones to expect a significant impact within three to five years and to be actively targeting these areas. (See Exhibit 3.) Strong innovators also show other signs of being focused on external innovation. For example, 75% report using incubators, 81% leverage academic partnerships, and 83% partner with other companies. Weak innovators lag consistently in all of these areas.

Platforms are technologies that provide a foundation for developing other business offerings. Numerous industrial goods companies, including Siemens (number 16) and Boeing (number 11), have built substantial platform businesses in predictive maintenance to complement their traditional engineering and manufacturing endeavors. Amazon, Microsoft, and IBM, among others, offer

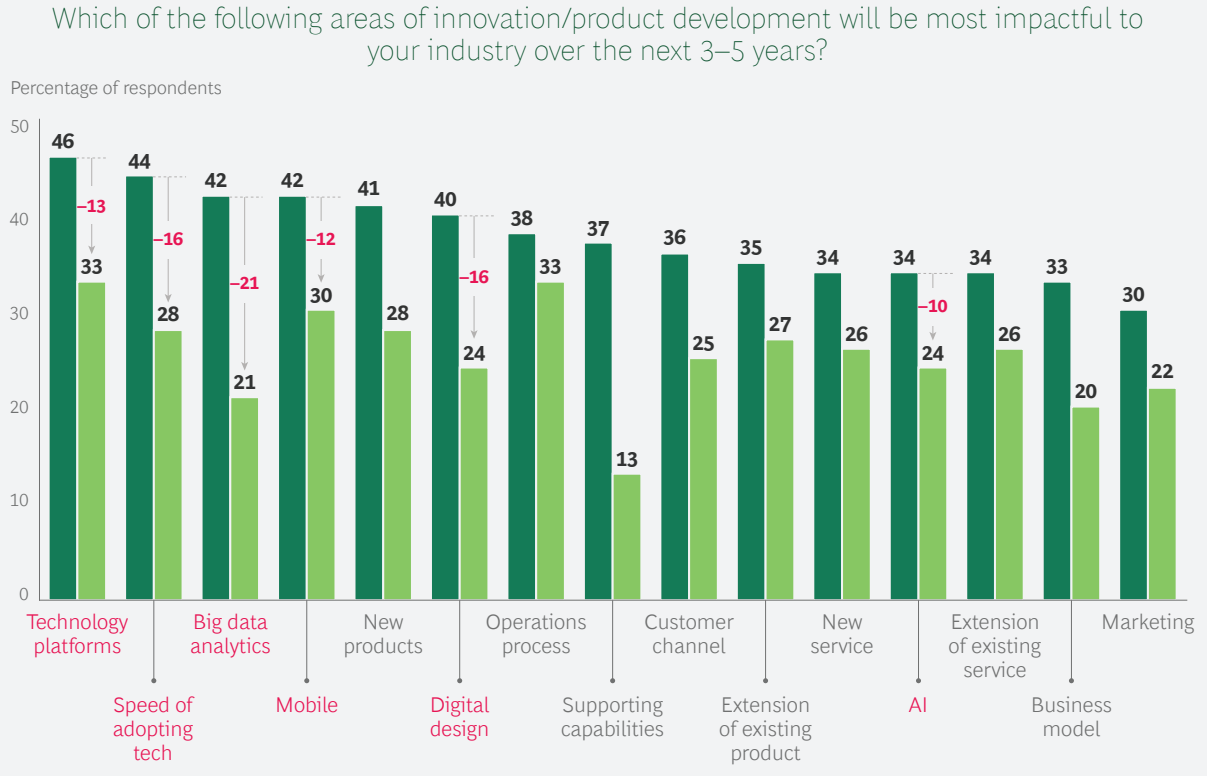
Exhibit 2 | Digital Innovations Are Among the Most Promising and Most Targeted Areas

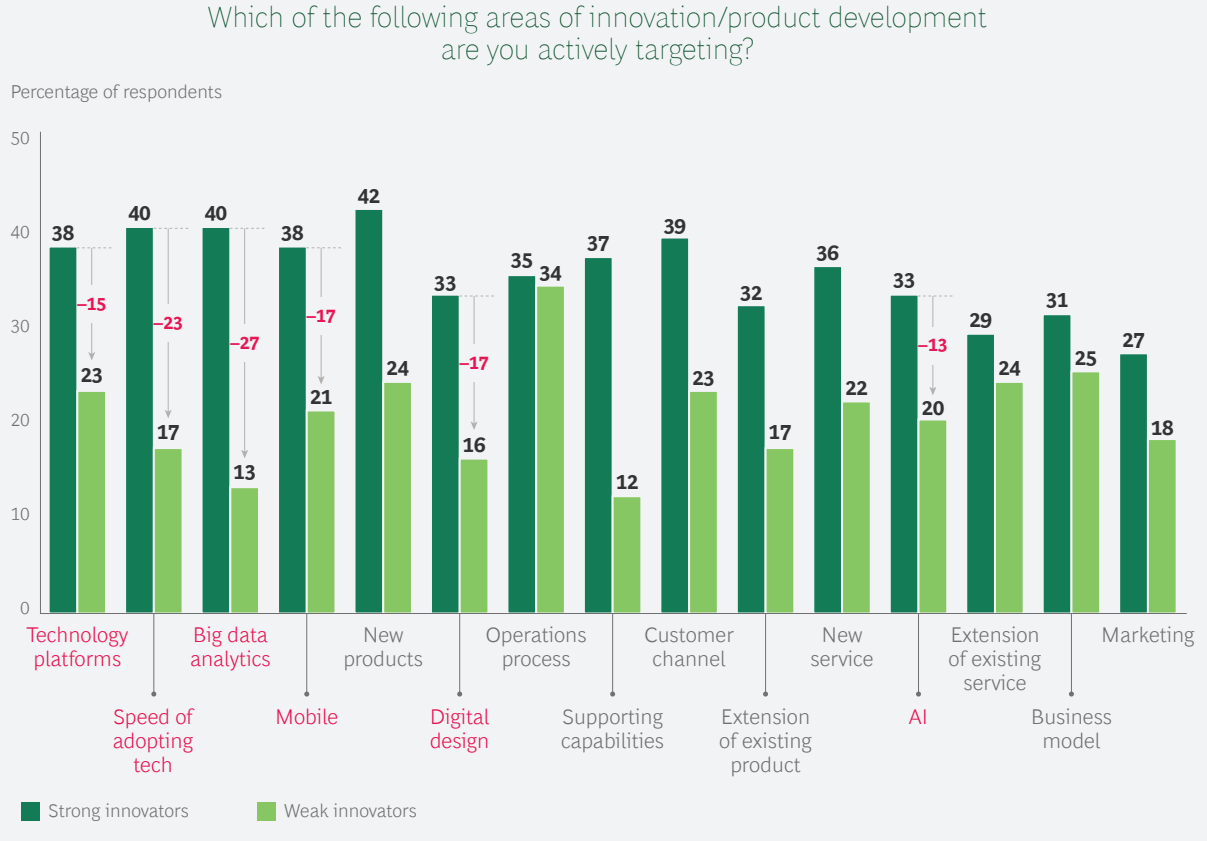




Sources: 2017 and 2018 BCG Global Innovation Survey.
Note: Areas identified with red labels are digital innovation categories.

Exhibit 3 | Strong Innovators Are Optimistic and Committed on Platforms and AI





Source: 2018 BCG Global Innovation Survey.
Note: Areas identified with red labels are digital innovation categories.

a range of software and services from their cloud platforms.

Ecosystems go a step further and leverage a range of partners that pull together the underlying technologies, applications, software platforms, and services needed to produce an integrated solution. (See "The Emerging Art of Ecosystem Management," BCG article, January 2019.) The two main mobile operating systems—Google's Android and Apple's iOS—have grown into complex ecosystems of telcos, device manufacturers, service providers, and application developers, among others. Rapidly changing technologies and growing customer demand for a highly customized user experience further amplify the need for partnerships.

The opportunity to innovate entirely new revenue streams, business models, and sources of continuing advantage is particularly strong for B2B businesses, thanks to the masses of data that devices connected to the Internet of Things (IoT) generate. Data ecosystems will play a critical role in defining the future of competition in many B2B industries. (See "How IoT Data Ecosystems Will Transform B2B Competition," BCG article, July 2018.)

The challenge is to set up and manage these ecosystems effectively and use them strategically to maximize value and gain a competitive edge. (For a look at how leading companies are building their own collaborative networks or joining existing ones to drive competitive advantage, see the third chapter, "How Platforms and Ecosystems Are Changing Innovation.")

One Additional Thought

A final observation—one that we have written about before—bears repeating, especially with regard to the four companies in this year's top ten list that were also in the top ten back in 2005. These companies are serial reinventors. Google constantly revises its algorithms and offerings. Amazon continuously invents new categories, services, models, and ways of engaging its customers. Microsoft has successfully transitioned from its longtime reliance on the Wintel partnership in PCs to become a cloud-based tech company. And IBM has reinvented itself multiple times, from mainframes to PCs to services to the cloud, AI, and platforms and ecosystems.

The tools and technologies of innovation evolve. The basic orientation toward change—never being satisfied and always being willing to reinvent oneself—remains part of some companies' lifeblood.

AI POWERS A NEW INNOVATION MACHINE

INNOVATION, MEET AUTOMATION. THIS could be the beginning of a beautiful friendship.

In a few short years, artificial intelligence (AI) and its subfield, machine learning (ML), have gone from futuristic vision to near-mainstream capability at many large companies, including in their innovation programs. A quick scan of BCG's 50 most innovative companies for 2019 shows that top innovators are also AI leaders (Google, Amazon, Apple, Microsoft, Netflix, and IBM, for openers) and that many others—including plenty of companies from traditionally nondigital industries (Boeing, Siemens, Marriott, BP, and multiple automakers, for example)—are actively leveraging AI.

Like any powerful new technology, AI is the subject of lots of hype. But in this case, real fire lies behind the smoke. The rapid ascent of AI in business is well chronicled in two reports by BCG and *MIT Sloan Management Review* (*SMR*).[1] The more recent of these reports found that AI leaders are deepening their commitment to the technology (as evidenced by funding) and seeking ways to apply AI at scale. Our current research into AI in innovation reveals similar trends. Nine out of every ten respondents to our innovation survey reported that their companies are investing in AI. More than 30% expect AI to be among the areas of innovation with the highest impact on their business in the next three to five years. And almost 30% are actively targeting AI in their innovation programs. Strong innovators are particularly active.

The 2018 *SMR*-BCG report also identified a performance gap associated with this critical new capability, indicating that AI "pioneers" appear to be "pulling further away" from their less aggressive peers. We see a similar pattern with respect to AI in innovation, which carries two key implications. First, AI is not a plug-and-play capability. Second, ML is inductive—that is, machines learn by doing, and they need to be fed large amounts of data to get smarter. Tortoises will have a tough time catching up with hares in this race. But hares can't afford to become complacent, because AI is truly disruptive and can propel long leaps forward.

What are the leaders doing?

A Strong Correlation

Our most recent innovation survey found a strong correlation between companies that consider themselves strong innovators and those that see themselves as being good at AI. (See Exhibit 4.) About 30% of respondents rate themselves as strong innovators, and about 25% as being better than average at AI. Almost 20% see themselves as both; we call this group "AI leaders." Nearly 17% of the respondents in our survey assess their orga-

EXHIBIT 4 | Nearly 20% of Companies Rate Themselves as Both Strong Innovators and Above Average at AI

Percentage of respondents identifying their company as:

	Weak innovator (%) 9.2	Average innovator (%) 61.3	Strong innovator (%) 29.5
Better than average at AI (%) 25.4	0.2	6.0	*AI leaders* 19.2
Average at AI (%) 58.1	1.9	46.6	9.6
Worse than average at AI (%) 16.5	7.1	*AI laggards* 8.7	0.7

Source: 2018 BCG Global Innovation Survey.

nizations as being below average on AI; we call this group "AI laggards." AI leaders are more likely than laggards to consider AI important to their organizations' future growth (94% versus 56%), which suggests that more than half of laggards will face a widening competitive disadvantage unless they step up their AI game.

AI leaders tend to see themselves as being good at embracing new technologies such as AI to enhance customer offerings or streamline the development process (89% versus 24%). AI leaders and laggards allocate their spending roughly equally among radical innovation; major new products, services, and internal process changes; and improvement of existing products, services, and processes.

But AI leaders are far more likely than laggards to apply big data and advanced analytics throughout the innovation process—from identifying new themes (91% for leaders, 30% for laggards) to informing investment decisions (92% to 27%). (See Exhibit 5.) In addition, by margins of 42% to 19%, and 42% to 24%, respectively, AI leaders are more likely than laggards to be actively pursuing the related avenues of big data and tech platforms.

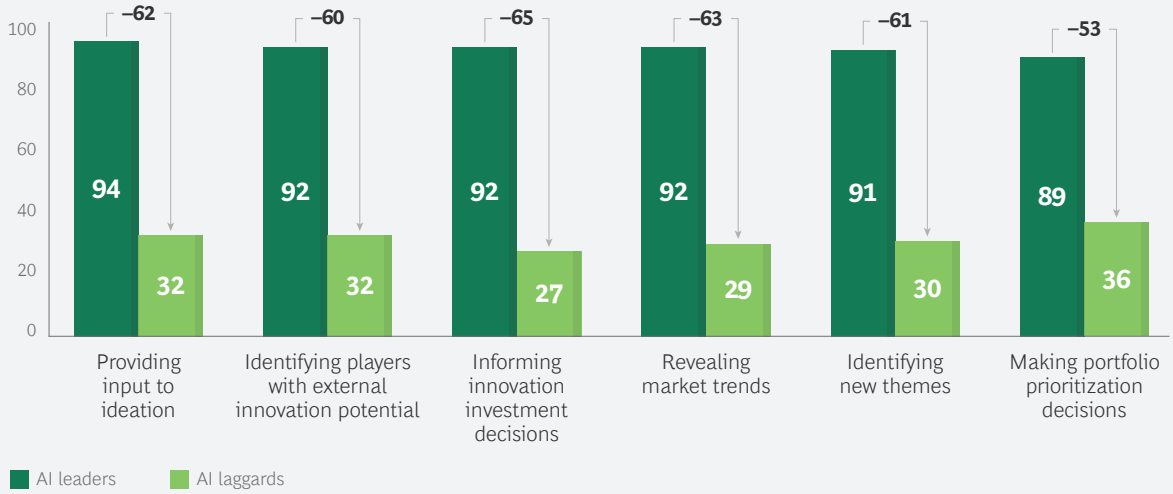
Where to Focus?

All companies face choices about where to apply resources and where AI might make the most impact. New examples emerge every day. BP (number 46 on the 2019 most innovative list) is using ML to improve its prediction models for oil and gas reservoir recovery. General Motors (number 40) used AI-aided design to reduce the weight of traditionally designed and manufactured parts by 40% while increasing their strength by 20%. Experts predict multiple areas of impact for AI in health care, including improving radiology diagnoses, making medtech devices smart, and identifying new infection patterns. In financial services, Chinese insurer Ping An, a digital native, developed predictive models and leveraged face and voice recognition as foundations of its business. The vice president and chief data scientist at Ant Financial, another Chinese digital financial services leader, told *SMR*, "AI is being used in almost every corner of Ant's business. We use it to optimize the business and to generate new products."

Early AI programs tend to focus on improving operational efficiencies, perhaps because companies can demonstrate success relatively quickly in these areas. The 2018 *SMR*-BCG re-

EXHIBIT 5 | AI Leaders Use Data Throughout the Innovation Process

How would you rate your company's skill at leveraging big data and advanced analytics to help with each of the following aspects of innovation?



Source: 2018 BCG Global Innovation Survey.

port found that many early AI initiatives were pilots or tests designed to solve a particular problem as companies focused their early attention on gathering low-hanging fruit and making moves with near-term impact. Nearly two-thirds of the AI leaders in our study report that their innovation-related use of AI aims to improve internal processes.

As significant as the impact of AI will be on business processes, however, its area of greatest potential lies in developing new products and services that can evolve into major revenue streams over time. Most of the AI pioneers in the *SMR*-BCG study prioritized revenue-enhancing applications, which is where 72% of respondents expect the big AI gains to originate. In our survey, the AI leaders reported much higher percentages of sales driven by AI-enhanced products or services introduced in the past three years: 46% of AI leaders say that 16% or more of sales are AI-generated, compared with only 10% of AI laggards). AI leaders also expect a much greater percentage of sales to come from the introduction of AI-enhanced products or services over the next five years (54% of AI leaders anticipate that 16% or more of sales will come from these sources, compared with 22% of AI laggards).

Despite ample evidence of impact, AI still garners its share of skepticism, which is one reason that laggards are falling behind. Research by BCG and Google in the consumer products sector found that by applying AI and advanced analytics at scale, consumer packaged goods (CPG) companies can generate more than 10% of their revenue growth through multiple means, including more predictive demand forecasting, more relevant local assortments, personalized consumer services and experiences, optimized marketing and promotion ROI, and faster innovation cycles. (See "Unlocking Growth in CPG with AI and Advanced Analytics," BCG article, October 2018.) Yet in our innovation survey, only 23% of CPG respondents said that they believe AI will have the greatest impact on innovation and product development in the industry over the next three to five years, and a similarly modest percentage are actively targeting AI in their innovation programs. Both figures are about 10 percentage points below the cross-industry averages.

AI at Scale

AI forces business executives to deal simultaneously with technology infrastructure and more traditional business issues. Typical IT

systems—which consist of data input, a tool, and data output—are relatively easy to modularize, encapsulate, and scale. But AI systems are not so simple. AI algorithms learn by ingesting data, and training data is an integral part of both the AI tool and the overall system. The entanglement is manageable during pilots and isolated uses but becomes exponentially more difficult to address as AI systems interact and build on one another. This leads to what our colleagues have termed the "AI paradox"—the ease of achieving powerful results with AI pilots and the difficulty of replicating those results at scale. (See *The Big Leap Toward AI at Scale*, BCG Focus, June 2018.)

Talent is a major issue for most organizations. Data scientists and software engineers aren't the only high-demand personnel. Companies also need people who combine business skills with an understanding of AI. Companies that are moving toward AI-centric innovation processes must consider how the transformation will affect their staffing in terms of skill shortages, job elimination, or both. In pharma, for example, as AI algorithms learn to identify the types of people and conditions that are best suited for clinical trials—a major process innovation—the need for human involvement in this critical function changes. Banks and other financial services companies are already feeling the impact of voice recognition and credit-scoring systems that use AI technology to automate customer service.

As AI becomes a more critical element of innovation, the talent challenge may widen the gap between strong and weak innovators. After all, Exhibit 4 indicates that more than 65% of strong innovators already see themselves as above average on AI, versus just 2% of weak innovators.

Build Versus Buy

One talent-related issue with immediate impact is the question of whether to build or buy AI capabilities. In our survey, 55% of AI leaders extensively use external vendors for AI projects—36% relying exclusively on them, and 19% relying mostly on them. This approach may help leaders climb the AI curve quickly, since expertise is still in short supply. But because of AI's heavy reliance on data, companies need to be careful about the agreements they construct with their suppliers and partners. They must protect their ownership of internal data, control how vendors use that data, and ensure their own continuing access to data from outside sources. (See "The Build-or-Buy Dilemma in AI," BCG article, January 2018.)

Companies need people who combine business skills with an understanding of AI.

AI leaders often rely on ecosystems. Digital natives, such as those leading our list of most innovative companies, grew up in ecosystems. Companies such as Amazon, IBM, and Microsoft offer AI capabilities for sale or rent through their cloud-based platforms. Some industrial companies—automakers and aircraft manufacturers, for example—are accustomed to working in broad partnerships of multiple companies, with all the collaboration and technology and data sharing that these arrangements entail. Others have less experience—but as we explore in the next chapter, "How Platforms and Ecosystems Are Changing Innovation," the fast-rising importance of technologies such as AI will all but force them to explore such alliances.

"Alexa, Innovate!"

A fascinating battle for AI-enabled smart-home systems is brewing among Amazon (Alexa), Google (Google Assistant), Apple (Siri), Microsoft (Cortana), and others as each company seeks to capture the biggest share of end-user households for its voice-recognition technology. The battle is a critical early-stage skirmish in a much longer conflict to determine how and for what purposes these systems will be used. At this point, streaming music is the most widely used application, but the larger competition may take a generation to decide. The plan appears to be for each company to open its smart-home platform to others, which will innovate actual use cases for consumer testing. Winners can be

scaled up quickly and losers tossed aside with equal dispatch.

Consider Alexa and Google Assistant. These classic platforms enable others to innovate inexpensively on top of them. By opening their platforms to others, Amazon and Google can harness others' innovations—much as Apple and Google did with their mobile platforms, and as Microsoft did with Windows. The platforms, provided at little or no cost to app developers, take care of the basic plumbing, and the platform owners derive value from having a richer set of services available than they would have had with their own apps alone. In this case, the stakes are high, as use of smart-home systems is expected to increase exponentially as today's young people grow up talking to, learning from, and coming to depend on these devices.

In China, Baidu uses a similar open-AI platform strategy in the B2B or industrial internet marketplace. Baidu provides access to AI services such as voice and image technology and natural-language processing, which companies are using in such diverse fields as agriculture, manufacturing, and health care, where these innovations have helped reduce the misdiagnosis rate for diabetes-related eye disease to less than 5%. (See "Get Ready for the Chinese Internet's Next Chapter," BCG article, January 2019.) Open platforms such as these will affect not only how consumers and businesses access AI applications but also how companies build their own AI capabilities.

It's still the early days in AI. But after years of development, actual use of the technology is catching on fast. Laggards need to get into the game, and leaders need to accelerate their efforts. The platforms offered by Amazon, Google, Baidu, and others may well make some kinds of experimentation and testing and learning easier. But the big issues of access to data and talent won't get any less challenging over time; if anything, competition for these vital assets will only increase. Companies that hope to rank among the most innovative of the future need to place their AI bets now.

Note

1. See S. Ransbotham, P. Gerbert, M. Reeves, D. Kiron, and M. Spira, *Artificial Intelligence in Business Gets Real*, *MIT Sloan Management Review* and Boston Consulting Group report, September 2018; and S. Ransbotham, P. Gerbert, M. Reeves, and D. Kiron, *Reshaping Business with Artificial Intelligence*, *MIT Sloan Management Review* and Boston Consulting Group report, September 2017.

HOW PLATFORMS AND ECOSYSTEMS ARE CHANGING INNOVATION

In 2016, a group of Finnish medical researchers won a global competition whose goal was to accurately predict prostate cancer survival rates. From an innovation perspective, two facts stand out. First, the competition was virtual: it was hosted and managed on a cloud-based platform that facilitates scientific collaboration. Second, before entering the contest, the Finnish team had not been active in cancer research.

More and more frequently, innovation today comes from outside the company.

As we noted in our 2016 report on the most innovative companies, innovation today can come from anywhere—and more and more frequently, it comes from outside the company. (See *The Most Innovative Companies 2016: Getting Past "Not Invented Here,"* BCG report, January 2017.) Technology has helped level the competitive playing field, making it easier for anyone anywhere to gain access to the hardware, software, and tools they need to develop new ideas and business models and connect with other people and organizations.

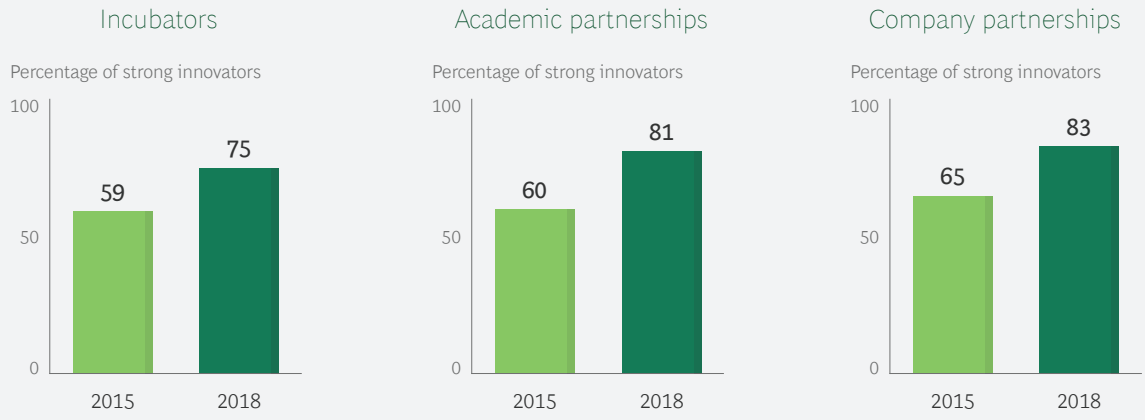
At the same time, companies can use an array of tools to cast wide innovation nets and get an early look at what's happening in their targeted fields of R&D. (See *How the Best Corporate Venturers Keep Getting Better*, BCG Focus, August 2018.) It's a powerful combination—if companies create the organization, processes, and incentives needed to bring outside ideas inside without killing them along the way.

It's an External Game

In business life today, companies need to look beyond their own walls for new ideas. We have seen big increases in the use of such partnership models as corporate venture capital, incubators, and accelerators. Our current innovation survey found that incubator use among self-described strong innovators rose from 59% in 2015 to 75% in 2018, that tapping into academic relationships jumped from 60% to 81% during the same period, and that formation of company partnerships increased from 65% to 83%. (See Exhibit 6.)

The increasing use of platforms such as Synapse, which hosted the prostate cancer challenge, is making innovation more multiparty and collaborative. There are many types of platforms—such as computing, technology, service, content, and marketplace—but all of them serve a similar purpose. Users access platforms because platforms combine aggregated supply or demand with high service levels and reduced friction. They can also facilitate collaboration and the development of

Exhibit 6 | Strong Innovators Are Increasingly Embracing Ideas from External Sources



Sources: 2015 and 2018 BCG Global Innovation Survey.

new ideas. For example, many of the longest-tenured companies on our most innovative companies list use platforms to gain access to different capabilities and sources of data, which they then use to build new business models or develop new products and services.

Developers of voice-recognition-based smart home platforms, including Amazon and Google, make it easy for others to create new consumer services that use their AI-enabled platforms—and in the process to attract the critical mass of applications needed to make their platform a clear leader.

Digital technologies enable collaboration platforms, and collaboration platforms enable ecosystems that bring together a group of organizations to build a new capability or product or service offering, or to help a new field of science or technology advance. Some ecosystems represent expansions of traditional ways of organizing and doing business; they tend to have an orchestrator at the center, with which all the other participants interact, along with established hierarchies and structures.

Other ecosystems—including many that are involved in the early research phase of R&D—tend to be more dynamic, less reliant on a central orchestrator, and more dependent on multifaceted interactions among participants. Various types of glue bind ecosystem participants. Money is one type of glue, of course, but knowledge, data, skills, and community can be equally important.

Just as multiple types of platforms exist, ecosystems may be formed to serve different purposes. Here are some of the most common ones that are relevant to innovation.

Building Capabilities. Many ecosystems are set up to bring in expertise from other industries. BCG research shows that 83% of digital ecosystems involve partners from four or more industries and 53% involve partners from six or more industries. For instance, a household electronics manufacturer might partner with sensor and camera manufacturers, an AI software provider, and research facilities to bring an advanced vacuum robot to market. In the past, to enter a new market (such as China), automakers either formed a joint venture or alliance with an OEM or established contractual relationships with hundreds of suppliers to secure parts. These traditional partnerships still exist, but today a typical auto company draws on an ecosystem of more than 30 partners across five different industries and several countries to make their cars connected, electric, and autonomous. The auto company acts as orchestrator, organizing and managing the ecosystem, defining the strategy, and identifying potential participants.

For example, Caterpillar's smart-mining ecosystem involves four types of collaboration, with a strong focus on flexible deal structures such as partnerships and minority investments, along with joint ventures and M&A. Caterpillar has forged partnerships to improve its proprietary industrial IoT platform's tech-

nology, and it has made minority investments to expand the platform's automation, optimization, and analytics capabilities. For most companies, these types of flexible deals require a different approval process than prevails in traditional contracting and M&A—one that is faster, leaner, and closer to the business. (See "The Emerging Art of Ecosystem Management," BCG article, January 2019.)

Developing New Products and Services. Ecosystems can offer multiple players a powerful way to build new revenue streams from products and services that they could not develop and bring to market on their own. For example, Alibaba's ecosystem provides a range of services for Alibaba platform users: travel, entertainment, gaming, finance, transportation, and e-commerce. It's an eclectic mix, and it generates a wide-ranging array of data that Alibaba collects and analyzes centrally. By capturing data from multiple sources, Alibaba can tailor individualized offers to users—timing them for maximum effectiveness—and provide tools that help its online sellers enhance their own businesses. The results, in turn, provide more data for the ecosystem.

In recent years, Alibaba has expanded on this model to build Ant Financial Services (originally Alipay) into a major force in financial technology; today, it is the world's most valuable fintech company, with more than 520 million users. Using machine learning technology and data from the Alibaba ecosystem, Ant provides an array of financial services to an underbanked market. Alibaba's data and analytics capabilities enable Ant to assign credit scores to individuals and small businesses, and the company has ventured from payments into online wealth management and consumer and small-business lending. In 2017, Ant launched two new initiatives: a payment option that uses facial recognition technology; and Ant Forest, which uses digital gaming technology to enable users to track their carbon footprint. The service has already attracted some 200 million users. (See "Digital Innovation on the World Stage," BCG article, May 2018.)

Collecting and Using Data. Data ecosystems will play a critical role in defining the future of competition, particularly in many B2B industries, because they enable companies to build data businesses. Such businesses are valuable not only because they generate recurring high-margin revenue streams, but also because they create competitive advantage. New data-driven products and services deliver unique value propositions that extend beyond a company's traditional hardware products, deepening customer relationships and raising barriers to entry by others. They also build highly defensible positions rooted in economies of scale and scope, similar to positions based on proprietary intellectual property (IP) or trade secrets. (See "How IoT Data Ecosystems Will Transform B2B Competition," BCG article, July 2018.)

Ecosystems enable players to develop new products and services collaboratively.

Gaining Access to IP. Shifts in the IP marketplace are creating new opportunities for companies that are long on IP to cooperate with traditional companies that may be short on it. Many tech companies, for example, have a surplus of IP, while industrial goods companies may face a deficit. In the IoT world, they can work together to expand the market for sensor-enabled and location-aware goods and services. Classic, exclusionary IP strategy is not dead, but companies need to master a more sophisticated and nuanced playbook for creating competitive advantage through IP. (See "The New IP Strategy: Make Love Not War," BCG article, June 2018.)

Merging Physical and Digital Channels. B2B and B2C companies are finding new ways to combine the digital world and the physical one to provide new services, more seamless customer experiences, and—in some cases—new business models. Amazon (with its acquisition of Whole Foods), Siemens (IoT), and the auto industry (connected cars and autonomous vehicles) offer examples. Emerging opportunities at the intersection of the digital and the physical give incumbents the chance to innovate and build valuable eco-

systems by defining new niches and new rules. (See "Getting Physical: The Rise of Hybrid Ecosystems," BCG article, September 2017.)

Advancing New Technologies. Companies that have a stake in basic scientific advances or in the development of new technologies such as quantum computing or synthetic biology may want to explore emerging deep-tech ecosystems—groups of entrepreneurs, startups, investors, corporate and academic partners, and others that join forces to research and develop a particular technology. Such technologies differ from others in their expected impact, in the time they take to reach market-ready maturity, and in the significant amount of capital (and investor patience) required to bring innovations to market. The ecosystems represent a new model that is far more fluid and dynamic than the methods used to conduct technology R&D in the past. (See *The Dawn of the Deep Tech Ecosystem*, BCG report, March 2019.)

Some Rules of Ecosystem Engagement

An ecosystem often operates differently from other types of business partnership, and in ways that can be foreign to big companies. Because ecosystems are dynamic environments that tend to contain lots of varied entities, they have multiple sources of influence, and no one party absolutely controls their direction. They trade in multiple currencies, not all of which are financial. Because they exist to create value through collaboration, they are the opposite of zero-sum congregations; when the ecosystem wins, each of the participants wins as well. And they evolve, acquiring new members and missions over time. (See the sidebar.)

NTT DOCOMO'S EVOLVING MOBILE ECOSYSTEM

Today it sounds like something from a distant era. In 1999—eight years before the launch of the iPhone—NTT Docomo (number 36 on the list of 2019's most innovative companies) built the first mobile internet ecosystem in Japan.

The leading telco developed a vertically integrated ecosystem based on partnerships and acquisitions that provided valuable services and experiences to feature phone users. Part of its plan was to drive "innovation through the convergence of mobile with other industries and services, thereby creating new values and markets." (See *Through the Mobile Looking Glass*, BCG Focus, April 2013.) Consumers flocked to new services such as mobile email, contactless payments, and, in time, streaming live video—all part of Docomo's "i-mode" ecosystem, which helped propel mobile phones to become Japan's most widely owned device and spurred traditional industries to go mobile. Docomo pioneered the technological innovation of mobile payments, leveraged local market dynamics, and achieved critical mass through partnerships with established players (often industry leaders). Establishing mobile payment capabilities for rail service helped seal other deals in the finance, retail, and hospitality sectors, among others, and helped attract more than 500,000 merchants to Docomo's mobile payment service.

Since the advent of smartphones, Docomo's ecosystem has undergone major evolutionary changes a couple of times. The first major change involved creating a mobile content hub through alliances with publishers, media companies, and other content producers. Docomo now offers more than 400 magazines online, as well as Japan's largest anime library. Its dTV service has a similar number of subscribers to Netflix in Japan. As the content business has become more competitive in recent years, Docomo has moved to the latest iteration of its ecosystem, which relies on its strengths in payments, user identification, customer loyalty, and other areas to offer promotional support and services to an expanding network of third-party e-commerce partners.

Companies that are considering orchestrating a new ecosystem or joining an existing one should take a number of steps in advance to prepare themselves:

- Make sure that you have a sharp, clear innovation strategy that identifies the innovation domains on which you will be concentrating.

- On the basis of your innovation strategy, clearly articulate an ecosystem strategy: what value you want get out of an ecosystem, and what you need to achieve in the ecosystem to support your broader innovation goals.

- Identify the types of players with which you will interact. These may extend beyond the obvious choices to companies in other (unrelated) industries, universities and research institutions, startups, and venture investors, among others.

- Think about how you will enlarge the ecosystem pie other than by simply maximizing value for your company alone. Consider how the ecosystem as a whole will create value and how the participants will share that value.

- Note the locations of current innovation hot spots around the world for the domains you are targeting. The US and China are primary hubs for many fields of new-technology R&D, but many other countries are active in particular fields and are developing serious centers of excellence. (See "A Deep Dive into Deep Tech Investing," BCG infographic, forthcoming.)

- Determine the kinds of relationships that will provide the foundation of the ecosystem, and decide on the nature of your involvement. For example, do you foresee a series of passive corporate venture-capital investments or active investments and collaborations? Is access to data more important than financial return?

- Align and manage your innovation vehicles (including their geographic locations, search patterns, guidance, and budgets), according to your answers to the preceding questions.

- Set clear goals, review achievements regularly, and don't be afraid to adjust your ecosystem strategy on the basis of real-world results.

The use of platforms and ecosystems will undoubtedly expand as more companies experiment with both types of vehicles and develop more successful use cases. The relentless advance of technology will also play a role. The more complex the technology, the narrower the expertise—and the more likely it will be that companies must look outside their own organizations for the skills needed to use the latest developments. But gaining access to the latest science is one thing. Bringing that science inside and integrating it with existing programs and processes is another. The latter remains the area where most companies will face their biggest challenges.

APPENDIX E

INSURANCE INDUSTRY

Redefining Global Health Insurance Through Value-Based Care

By Matthias Becker, Nicolas Busch, and Kerstin Helmlinger

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It's a challenging time for health insurance companies around the world. Private health insurers—whether they have operations in one or multiple countries—face increasing pressure as global healthcare costs continue to rise and profit margins shrink. While patients demand superior services, they are dissatisfied with rising premiums. At the same time, despite significant insurer

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investments in new products and services, improvements in patient outcomes aren't keeping pace with healthcare cost increases. This dilemma has given rise to numerous discussions between BCG and leading health insurance executives about how to strategically address these pressures.

There are many reasons for rising costs, some of them beyond insurers' control. An aging population uses more medical services, while newer and more costly treatments, many of them pricey new drugs, continue to emerge. Many medical advances have revolutionized treatments in recent years, yet they are not always more effective. Overall, a quarter of the treatments patients receive are considered unnecessary, adding needless cost while doing little or nothing to improve health outcomes.

To rein in spending and improve outcomes, we believe that insurers should look to a strategy that, while not new, has yet to reach its full potential: value-based healthcare . Rather than focusing on the quantity of services provided, this approach emphasizes clinical value and best outcomes. It aligns patient, provider, and insurer financial incentives to encourage the use of high-value care and discourage care that is low value.

Our approach to value-based care consists of five levers aimed at both insurance policyholders and providers that together can help health insurers begin to implement these practices. The goal, no matter an insurer's size or market share, is to control rising costs, improve healthcare outcomes, and increase customer satisfaction simultaneously. In this way, insurers can redefine the role of health insurance in promoting good health and truly go from payer to player by focusing on what can be achieved through value-based care rather than chasing reimbursement on a per-service basis.

Market Growth Has Failed to Drive Better Outcomes

In recent years, the global health insurance market has grown substantially. In 2024, it is projected to reach $2.42 trillion in gross written premiums. Overall, the gross written premium compound annual growth rate from 2024 to 2029 is projected by Statista to be 3.18%, resulting in a market volume of $2.83 trillion by 2029.

Historically, the market has experienced premium growth rates of roughly 5% annually. Going forward, we expect that there may be even more marked increases in global gross written premiums, reaching a maximum estimated 10% annual growth. This would bring the global market size to more than $4 trillion by 2029. (See Exhibit 1.)

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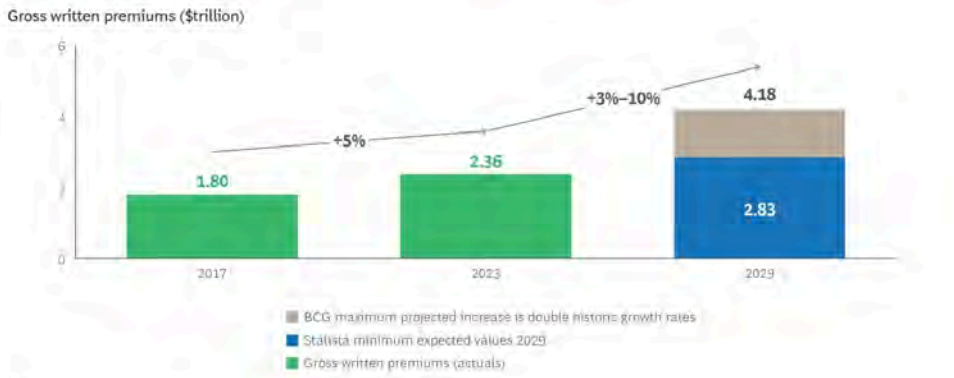
Exhibit 1 - Global Premiums Have Increased Strongly and Are Projected to Increase Further



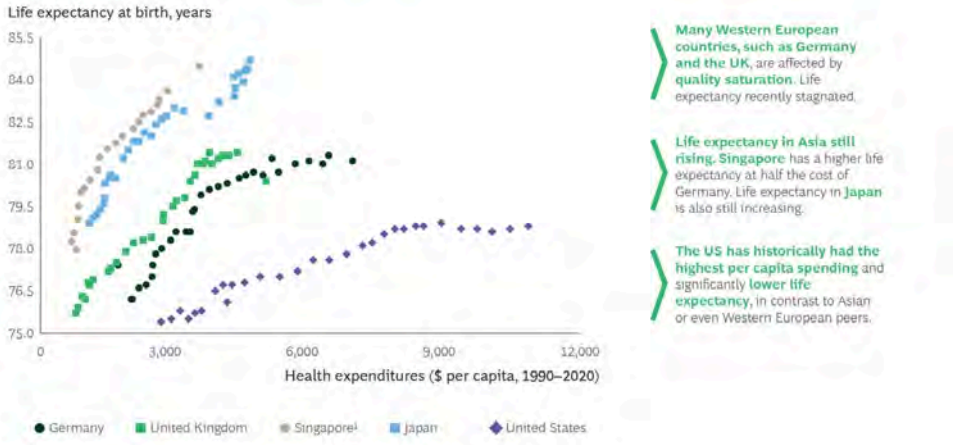
Sources: Statista; BCG analysis.

Profit margins have remained relatively constant over time. But pressures on health insurance companies are mounting. For example, private sector insurance premiums in many countries are at an all-time high in 2024 due to insurers covering higher healthcare costs in recent years.

At the same time, despite substantial annual premium increases, patient outcomes have not improved commensurately, and life expectancy worldwide has barely budged in several years. The gap between rising healthcare costs and stagnant life expectancy can be attributed to several factors, including lack of access to care, costs associated with managing chronic conditions, and a focus on treatment rather than prevention, as well as the aging population. These complexities mean that simply spending more on healthcare does not guarantee better outcomes or longer life expectancy.

Looking at the correlation between life expectancy and general health expenditures, we observe that many leading European countries and the US seem to be approaching quality saturation, where higher spending leads to diminishing improvements in health. Some countries such as Singapore, however, continue to show improving life expectancies despite comparatively lower health expenditures. (See Exhibit 2.)

Exhibit 2 - In Many Western Countries, Life Expectancy Is No Longer Rising Even as Health Care Costs Increase



Sources: Organization for Economic Cooperation and Development and World Health Organization data.
[1]Data from 2000–2020.

Value-Based Care Can Help Insurers Cut Costs While Improving Patient Outcomes

The concept of value-based healthcare was introduced more than 15 years ago by US business strategy and innovation experts Michael Porter and Elizabeth Teisberg. At its core, value-based care provides financial incentives to patients, providers, and insurers to choose care that has greater clinical value and avoid low-value care.

While the model has gained some traction, primarily among doctors and healthcare systems, insurers have faced several barriers to adopting this approach. Notably, the complexity of transitioning to a value-based model from fee-for-service payment systems, provider resistance, patient unfamiliarity with how the model affects their care, and inadequate data infrastructure have proven difficult for insurers to address. Some insurers have also expressed reservations about whether their comparatively small market share gives them enough power to seriously change practices of healthcare systems they service.

We believe there are important elements of value-based care that health insurers of any size can leverage to bring higher quality care and lower costs to their members. Despite the particularities of different healthcare systems, insurers are very well positioned to drive value-based care and lay the groundwork for a larger transformation of these systems. (See Exhibit 3.)

Exhibit 3 - A Value-Based Care Framework for Health Insurers

Source: BCG analysis.

Key Actions for Insurers to Implement Value-Based Care

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Principles

BCG's approach consists of five interlocking levers that health insurers can use as a framework to begin to advance the practice of value-based care within healthcare systems. These levers apply technology, policy, and data reporting to raise the profile—and the benefits—of value-based care in the health insurer ecosystem.

Develop highly individualized tools to guide policyholder choices.

Today, many of the interactions that health insurers have with policyholders are not focused on their needs. By developing targeted, individualized communications and digital tools, insurers can change members' expectations and gain their trust on questions unrelated to insurance reimbursement. At the same time, this shift in focus can differentiate insurers from other payers and provide a competitive advantage.

To implement this change, insurers should update products to encourage value-based behavior by policyholders. This can include preferred access to providers that balance quality and cost as well as higher co-payments for choosing non-preferred providers or out-of-network hospitals. Policyholders also can be offered incentives for receiving preventive care.

Engage policyholders with an integrated digital portal.

When policyholders have questions about their health and care that are unrelated to coverage, health insurers are typically not the first partner they turn to for advice. To encourage value-based care behavior in policyholders, insurers need to become the focal point for all of their health-related questions. By implementing easy-to-access and easy-to-use digital portals, insurers are better positioned to influence consumers' health choices.

Establish value-focused provider networks.

Some countries have a standardized framework for what hospitals are allowed to charge and which treatments they can offer; however, that is not the case in many developing countries. Providers have considerable influence in these countries, including the freedom to determine the exact method of treatment and the rates to charge insurers. Within this framework, by utilizing modern analytics and stringent provider management, health insurers can successfully negotiate rates,

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develop treatment packages, and have clear governance in place to ensure the highest quality clinical care at an acceptable cost. Even if reimbursement rates are fixed, insurers can still benefit by referring their clients to top providers with the best medical outcomes.

Promote quality measurement and reporting.

Increasingly, health insurance companies have become interested in measuring and reporting on quality of care. Payers see a new role for themselves to change the system proactively by leveraging value-based incentive models. By tracking and analyzing additional data points and reporting them to hospitals, they are also putting in place the foundation for value-based reimbursement. The main issues insurers face in promoting and using quality data is the fragmented provider sector and the lack of quality data about outcomes.

Provide care coordination to help policyholders navigate the healthcare system.

Care coordination solutions focus on addressing policyholder issues within the healthcare system, particularly among different providers. Through care coordination, insurers can guide policyholders through their healthcare "journey" and help them with selecting quality providers while managing treatment costs.

Frequently, insurers are worried that they cannot influence, let alone steer, their policyholders' health decisions. The implementation of care coordination gives insurers an opportunity to establish trust with policyholders regarding their wellbeing. Therefore, insurers need to build up capabilities to advise and help policyholders navigate the provider ecosystem and select quality specialist services.

Key Technology Investments Fuel Successful Implementation of Value-Based Care

To address mounting cost pressures, health insurers must prioritize the implementation of value-based care. Technology is crucial in these efforts, including adopting generative AI (GenAI) and

developing data and analytics capabilities. Without sufficient investment in technology, health insurers will continue to operate programs in silos with limited cross-collaboration and with a typically reactive mindset that fails to generate actionable insights. Concerns that a single insurer is too small or not powerful enough to implement the value-based care model should no longer be a limiting factor with the help of new tools. Two actions will be key to successful implementation.

Develop a cutting-edge data and analytics platform.

As healthcare costs rise and governments or consumer demand limit insurance premium increases, insurers are increasingly interested in using data and analytics more systematically. There is an ever-increasing need to drive faster, more proactive insights; increase transparency across programs; and better leverage technology to accelerate the rate of learning across insurance organizations. Building the foundational technological capabilities, including an augmented set of dashboards that leverage AI to generate cross-cutting insights, has become a necessity.

Adopt new technologies like GenAI that promote individualized interaction.

In the past, many interactions with policyholders used standardized materials and approaches because tailoring information was prohibitively expensive. But with the help of new technologies, individualized interactions can be affordable to design. “Digital first” is becoming the dominant new approach at companies, and prioritizing digital channels helps address many pain points for policyholders—such as access to medical records and examination results and long wait times for appointments—while facilitating provider interactions with insurers.

While the value-based-care transformation of the world’s healthcare systems is just beginning, these steps will enable health insurance companies to position themselves for its successful adoption, improving their costs in the short term and providing policyholders with the best possible care in the long term. We believe that even smaller insurers can benefit from the levers of value-based healthcare outlined in this article. They can control rising costs, increase customer satisfaction, and offer themselves as a true healthcare player rather than “just” a payer. Importantly, insurers don’t need a dominant market share to benefit from these strategies, but moving toward value-based approaches can improve their competitive position very early on.

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